LM01 Firms and Market Structures

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1. Introduction

This learning module covers:

- Breakeven and shut down points of production, and economies and diseconomies of scale.
- Characteristics, demand, supply, optimal price, and output for different types of market structures: perfect competition, monopolistic competition, oligopoly, and pure monopoly.
- Techniques used by analysts to identify what market structure a firm is operating in.

2. Profit Maximization: Production Breakeven, Shutdown and Economies of Scale

Firms can generally be classified as operating in either a perfectly competitive or imperfectly competitive environment.

In a perfectly competitive market,

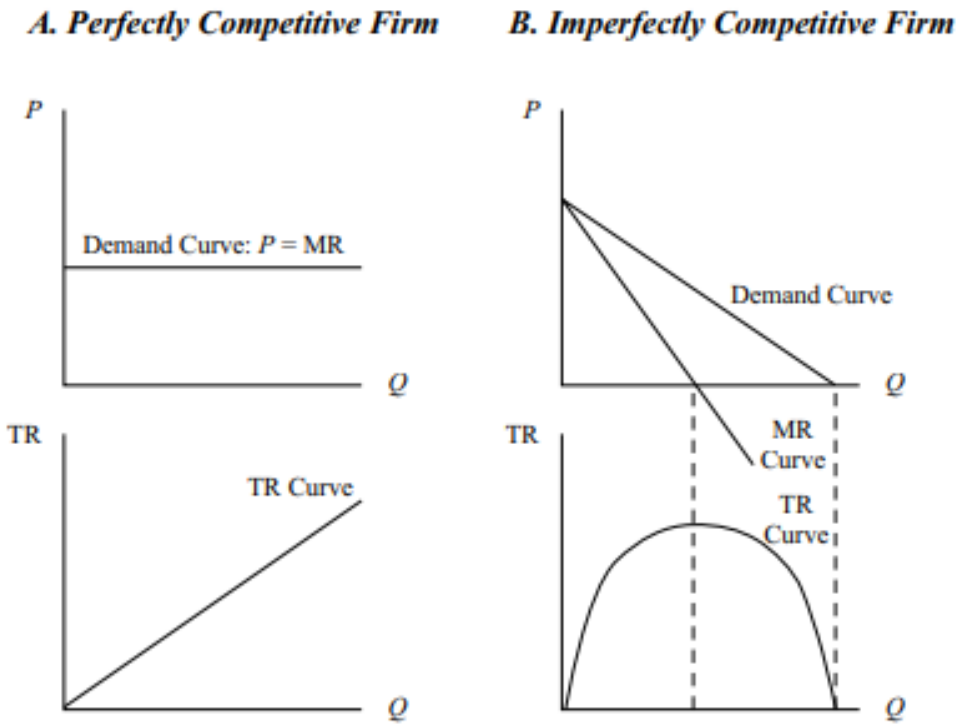
- The firm faces a perfectly elastic, horizontal demand curve; and it must take the market price of its output as a given. It has no pricing power.
- Its marginal revenue (MR) = average revenue (AR) = price of the product (P).
- Its total revenue (TR) = P x Q. As the firm sells one more unit its TR rises by the exact amount of price per unit.

In an imperfectly competitive market,

- The firm faces a downward sloping demand curve. It can set prices, but if it sets a high price the quantity sold is low, to increase sales it has to lower prices.
- MR curve is also downward sloping and lower than the demand curve. It intersects the X-axis at the point where total revenue is maximized.
- The TR curve for such a firm is initially zero, then it increases and subsequently decreases. It increases when MR is positive and demand is elastic. It falls when MR is negative and demand is inelastic. TR is maximum when MR is zero.
- Total revenue increases with greater quantity. However, there is a quantity at which the profit is maximized. Beyond this, any price decrease will result in a decrease in total revenue as the effect of the decrease in price will be greater than the quantity sold.

Exhibit 1 from the curriculum illustrates these concepts.

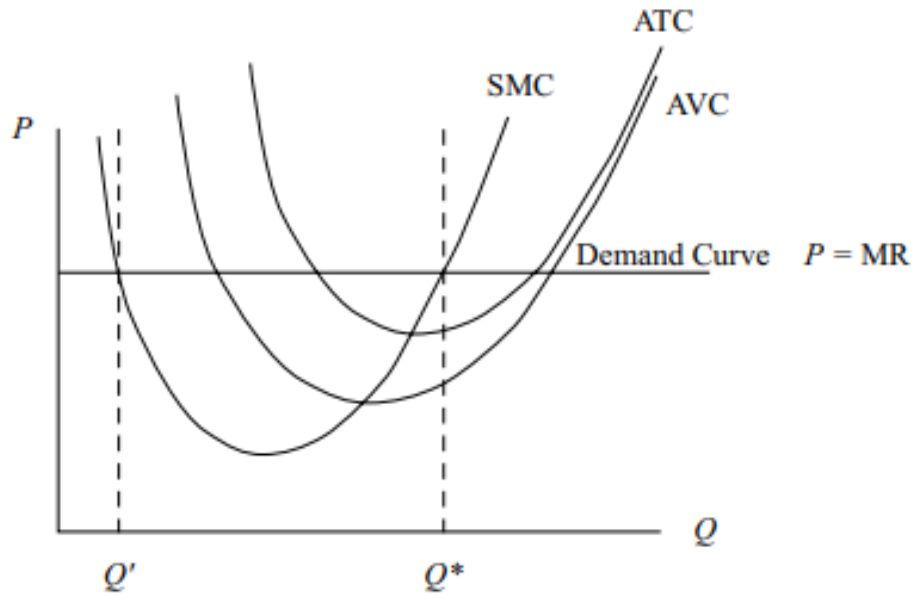
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Profit-Maximization, Breakeven, and Shutdown Points of Production

We now combine the firm's short-run TC (total cost) curves and TR curves to represent profit maximization under both perfect and imperfect competition.

Exhibit 2 from the curriculum shows the profit maximation conditions under perfect competition.



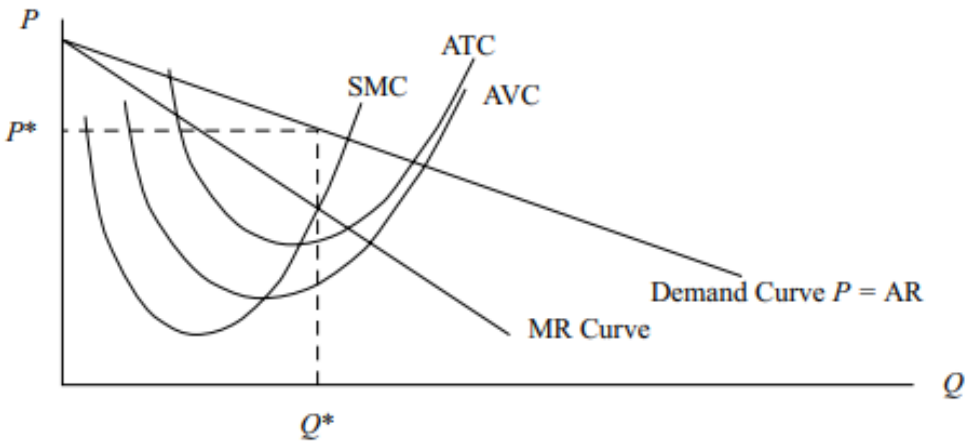
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Profit is maximized by producing Q*, where MR is equal to the short-run marginal cost (SMC) and SMC is rising. Note that MR is also equal to SMC at Q', but here the SMC is falling, so this is not the profit-maximizing output level.

If the market price rises, the firm's demand and MR curve will shift up, and the new profit-maximizing output level will be to the right of Q*. Similarly, if the market price falls, the firm's demand and MR curve will shift down, and the new profit-maximizing output level will be to the left of Q*.

As shown in the diagram, the firm is currently earning a positive economic profit because the market price is above the average total cost (ATC). This scenario is possible in the short run; in the long run, however, competitors will enter the market to capture some of those profits, and the market price will be driven down to level equal to each firm's ATC.

Exhibit 3 from the curriculum shows the profit maximization conditions under imperfect competition (a monopoly).



Although the MR and demand curves are different, the profit maximization rule stays the same: Find the optimal output level Q* where MR = SMC. Once this optimal output level is found, the optimal price P* is given by the firm's demand curve.

The monopolist earns positive economic profit because its price exceeds ATC. Since a monopolistic market usually has barriers to entry that prevent competition, the firm can continue to earn positive economic profits in the long run.

Breakeven Analysis and Shutdown Decision

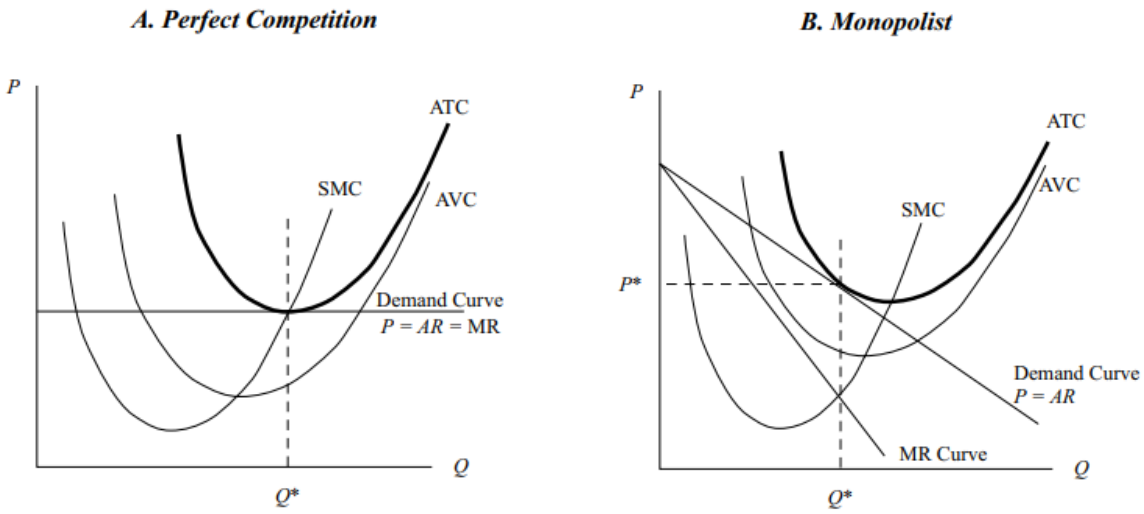
A firm is said to break even under the following conditions:

- total revenue equals total costs (TR = TC).
- price (average revenue) equals average total costs (AR = ATC).

When a firm is operating at its break-even point, the economic profit is zero. It might still be earning a positive accounting profit.

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Exhibit 4 from the curriculum depicts the breakeven scenarios for both perfect competition and imperfect competition (a monopoly).



In both scenarios, at the optimal output level Q* (where MR = SMC), the price is equal to the ATC. Hence economic profit = 0 and the firms are breaking even.

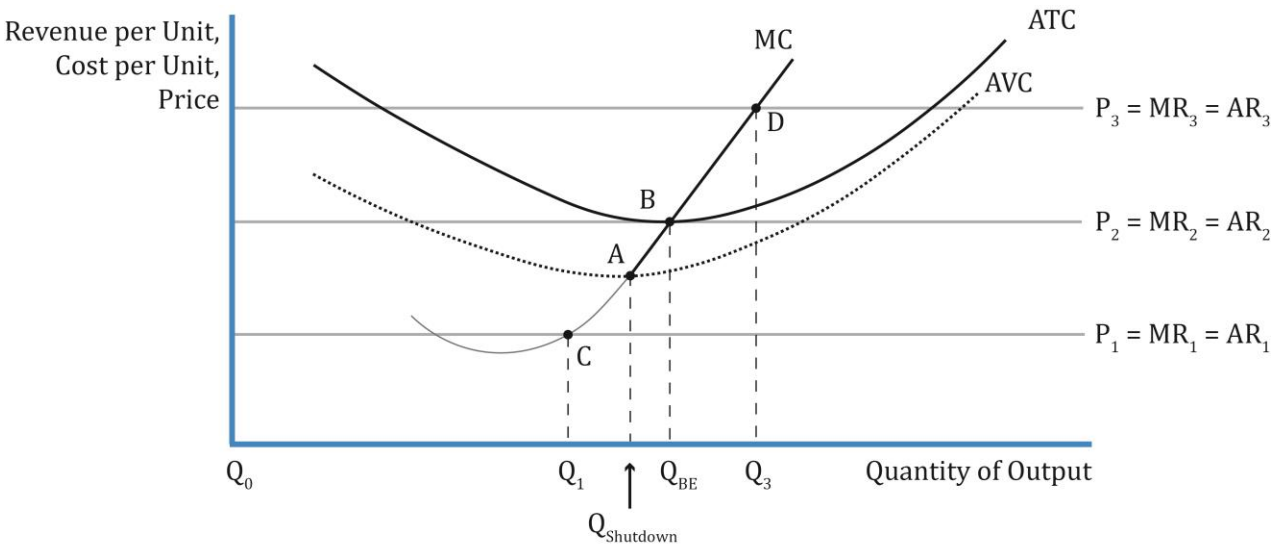
The Shutdown Decision

The relationships that show when a firm must operate or shutdown are given in the table below:

Short-run effect of the relationship between price and ATC on a firm		
Situation	**Short run**	**Long run**
TR ≥ TC	Operate	Operate
TR ≥ TVC but TR < TC	Operate	Exit
TR < TVC	Shutdown	Exit

Let us understand a firm's breakeven and shutdown point using the graph below.

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Interpretation of the graph:

- Assume the price at P_3 is 150. If the competition is perfect, then P_3 is the demand curve and MR = AR.
- At any point on the MC between P_2 and P_3, the firm is profitable because the average revenue is greater than the average total cost.
- The breakeven point is the point where P = ATC = MC. Graphically, it is the point where MC intersects ATC. The corresponding quantity is the breakeven quantity, Q_{BE}. Suppose this price is 100.
- Between A and B, the price is greater than AVC. The firm will continue to operate in the short run even though it is not profitable.
- To the left of A, the price is less than AVC. The firm will shut down.

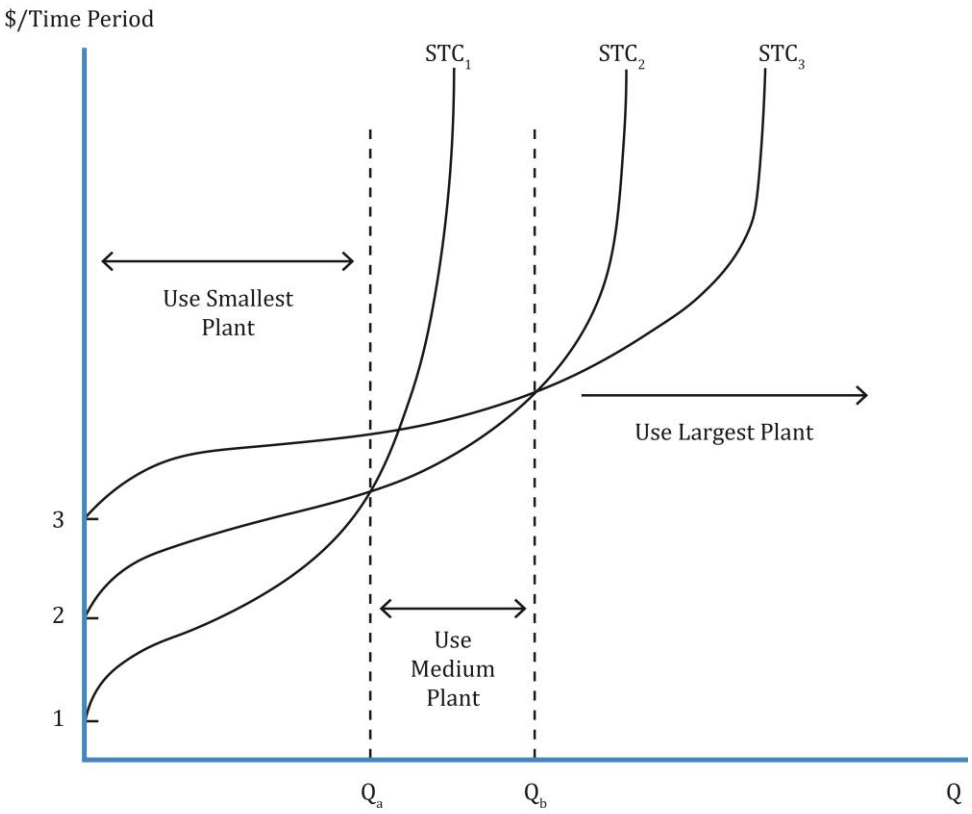
Economies and Diseconomies of Scale with Short-Run and Long-Run Cost Analysis

Economists use two time horizons based on how firms are able to vary the quantity of input: short run and long run. In the short run, at least one of the inputs or factors of production is constant. In the long run, all factors of production are variable.

Short- and Long-Run Cost Curves

The graph below shows a set of short-run total cost curves for each level of capital input.

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In the short run:

- Typically, where we are considering two factors of production, capital is fixed in the short run and the variable factor is labor. However, when capital changes, we get a new STC curve for each level of capital input.
- The fixed-input constraint along with other input prices determines a firm's short-run total cost curve (STC).

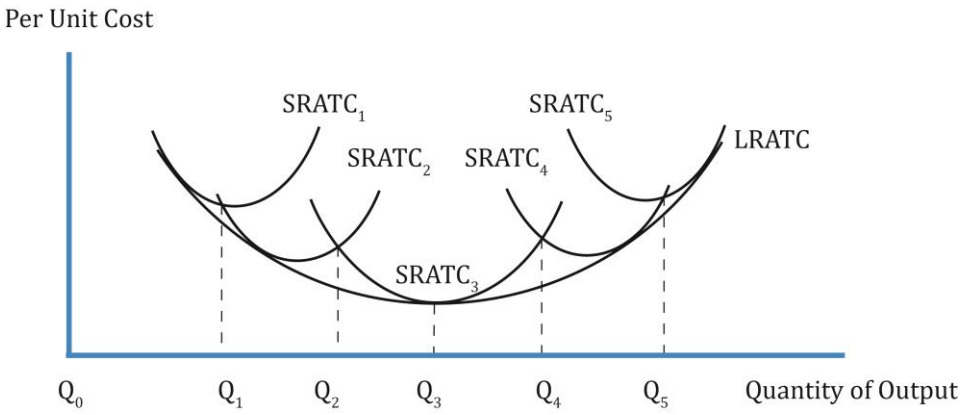
In the long run:

- All factors of production (both labor and capital) are variable.
- Think of the long-run total cost curve (LTC) as a combination of several STCs. By drawing a tangent to the minimum point of all the SRATC curves and connecting them, we get the LTC curve for the firm.
- The LTC is called the envelope curve. It envelops or encompasses all combinations of technology, plant size, and physical capital.

Defining Economies of Scale and Diseconomies of Scale

Each STC curve has a corresponding short-run average total cost curve (SRATC). The STCs for different plant sizes and the corresponding long-run average total cost curve (LRATC) is shown in the exhibit below.

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Interpretation of the graph:

- The SRATC defines what the per-unit cost will be for any quantity in the short run.
- The SRATC shifts down and to the right. Note that as plant size increases, the per-unit cost decreases as can be seen in the case of $SRATC_3$.
- The LRATC is derived from connecting the lowest level of STC for each level of output.
- The shape of the long-run cost curve depends on whether the firm is facing economies of scale or diseconomies of scale.
- **Economies of scale** is the decrease in the long-run cost per unit as output increases. LRATC has a negative slope when there are economies of scale. The portion to the left of Q_3 represents economies of scale.
- Q_3 represents the **minimum efficient scale**. It is the output level at which the long-run average total cost is the lowest and the output is optimal. This portion exhibits constant returns to scale where long-run average total costs do not change as output quantity increases.
- Beyond Q_3, the LRATC goes up. This portion represents **diseconomies of scale**. Here there is an increase in long-run cost per unit as output increases. LRATC has a positive slope when there are diseconomies of scale. The right side of LRATC curve represents diseconomies of scale.

The factors contributing to economies of scale and a lower ATC are as follows:

- Increasing returns to scale: increase in output is relatively larger than the increase in input. The left side of Q_3 shows increasing returns to scale.
- Division of labor/management.
- Technologically/economically efficient equipment that results in increased productivity.
- Effective decision-making.
- Reduce waste and lowering costs through better quality control.
- Bulk purchases resulting in lower prices.

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The factors contributing to diseconomies of scale are as follows:

- Decreasing returns to scale: Increase in output is relatively less than the increase in input. The right side of Q_3 shows decreasing returns to scale.
- Higher resource costs due to supply bottlenecks.
- Improper management because of size.
- Duplication of product lines.
- Higher labor costs.

3. Introduction to Market Structures

Analysis of Market Structures

The market is defined as a group of buyers and sellers that are aware of each other, and are able to agree on a price for the exchange of goods and services.

The market structure is classified into the following four categories:

- Perfect competition
- Monopolistic competition
- Oligopoly
- Monopoly

Perfect competition and monopoly are two extremes of the market structure in terms of number of firms and profits with the other types falling somewhere in between.

Factors that Determine Market Structure

The five factors that determine market structure are:

- The number and relative size of firms supplying the product. The higher the number of firms, the higher the degree of competition.
- The degree of product differentiation.
- Pricing power of the sellers. Are they price takers, or can they influence market prices?
- The relative strength of the barriers to market entry and exit.
- The degree of non-price competition.

The table below summarizes the basic characteristics of the four market structures:

	Perfect Competition	**Monopolistic Competition**	**Oligopoly**	**Monopoly**
Number of Sellers	Many firms	Many firms	Few firms	Single firm
Barriers to Entry and Exit	Very low	Low	High	Very high

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Product Differentiation	Homogeneous	Substitutes but differentiated	Close substitutes or differentiated	Unique product
Non-price Competition	None	Advertising and product differentiation	Advertising and product differentiation	Advertising
Pricing Power	None. Price taker.	Some	Some to significant	Considerable
Example	Oranges; Milk; Wheat	Toothpaste	Prices of commercial airlines for a given route	Electricity provider/any utility company (water, cooking gas) as they are typically controlled by a government authority

Note: This table is important from an exam perspective.

The most preferred market structure by producers is monopoly/oligopoly because they offer the highest pricing power. The most preferred market structure by consumers is perfect competition as prices are lower.

4. Monopolistic Competition

This is a market where there are many sellers of slightly differentiated products. *Product differentiation* is the key here. Ex: Burgers sold by KFC, McDonalds, Burger King, etc.

If the firm is able to successfully differentiate the product (e.g. Harley Davidson motorcycles), then the firm acts like a small monopoly.

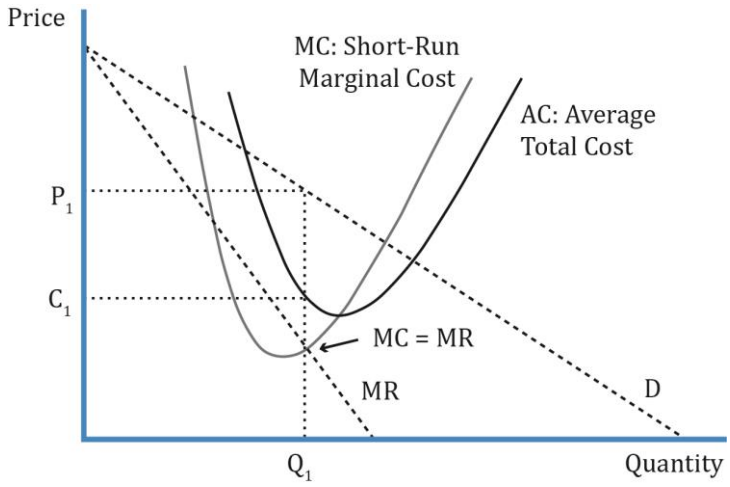
Characteristics**:**

- There are a large number of potential buyers and sellers.
- The products offered by each seller are close substitutes for the products offered by other firms, and each firm tries to make its product look different.
- Entry into and exit from the market is possible at fairly low costs.
- Firms have some pricing power.
- Firms differentiate their products through advertising and other non-price strategies.

Demand Analysis in Monopolistically Competitive Markets

The graph below shows the marginal cost (MC), demand and marginal revenue (MR) curve for a monopolistic firm.

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- Since the products are unique, a monopolistic firm has a downward sloping demand curve. MR is steeper and lies below the demand curve.
- Let us consider the toothpaste market. If consumers believe using Sensodyne toothpaste will relieve them of toothaches, then they will buy the product. However, the firm will have a downward sloping curve because if the prices are very high, then consumers will not buy the product, and will look for alternatives. Conversely, demand increases when the price decreases.
- Price and quantity demanded are inversely related.
- In the short run, the profit-maximizing quantity is MR = MC. This is Q_1 in the graph. The price is then determined based on the demand curve. This is P_1 in the graph.
- Because the product is differentiated, firms have some pricing power and charge what is determined by the demand curve. But each time a new firm enters the market, the demand curves of other firms fall (i.e., they lose a part of the market share). Since there is high competition, the products are often priced closed to each other.
- Demand is elastic at higher prices and inelastic at lower prices.
- Total revenue = $P_1 * Q_1$
- Cost = $C_1 * Q_1$
- In the short run, economic profit = $(P_1 * Q_1) – (C_1 * Q_1)$

Supply Analysis in Monopolistically Competitive Markets

Optimal Price and Output in Monopolistically Competitive Markets

Key points related to supply analysis in the context of monopolistic completion are:

- Output is based on MR = MC.
- Price is determined based on the demand curve.
- The supply function is not well-defined in monopolistic competition.
  - Recall that in perfect competition, a firm's output does not affect the price as all units sell at the same price (horizontal demand curve). They are price takers. P =

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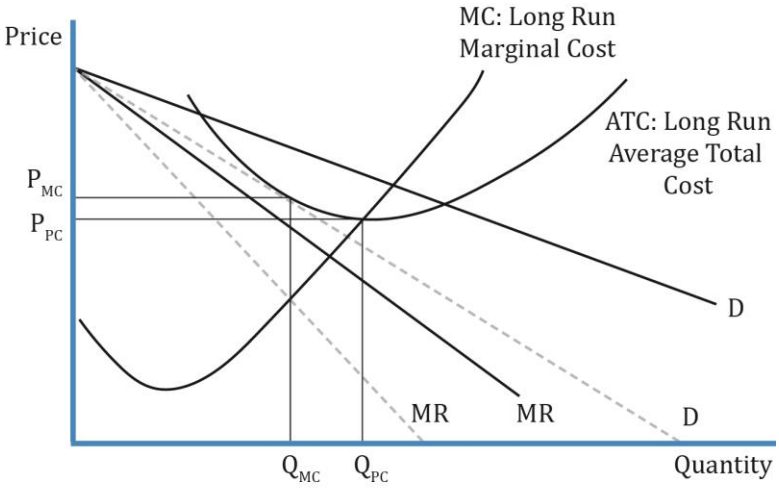
MR = MC. But, how much a firm produced was dependent on its MC curve. In the short run, the firm's supply curve was the MC curve above the minimum point of the AVC curve. The MR curve was a flat line and the same as the market price at that point.

- But, in monopolistic competition there is no single price as the firm can set its own price, and does not have to take the price determined by the market. The price here is determined by the demand curve. The firm's supply curve must show the quantity the firm is willing to supply at various prices, which is not shown by the MC curve here. MR is a downward sloping curve. The optimal output is still the intersection of MR and MC.

- Prices are higher and quantity is lower relative to perfect competition.
- Total profit = TR – TC.

Long-Run Equilibrium in Monopolistic Competition

The graph below shows the long-run marginal cost (MC), long-run average cost (ATC), demand and marginal revenue (MR) curve for a monopolistic firm.



Interpretation of the graph:

- The solid lines show the original demand and MR curves.
- The dashed lines show the shift in the demand and MR curves when a new firm enters the market.
- Short-run economic profits of existing firms encourage new firms to enter the market as the barriers to entry are low. When new firms enter, the demand curve shifts to the left and the market share of existing firms falls. The number of products in the market increases.
- In the long run, firms will enter and exit until P = ATC. At this point, economic profit will be zero and there will no longer be an incentive for new firms to enter the market. Therefore, long-run equilibrium is established.

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- Q_{MC} and P_{MC} are the equilibrium quantity and price respectively, for monopolistic competition. Q_{PC} and P_{PC} are the equilibrium quantity and price respectively, for perfect competition. As you can see, the equilibrium price is higher and the quantity is lower for monopolistic competition.

The table below summarizes the similarities and differences between perfect competition and monopolistic competition.

Perfect Competition (PC) vs. Monopolistic Competition (MC)	
Similarities	**Ways in which MC is different from PC.**
Long-run economic profit is zero	In the long run, profit-maximizing output quantity of MC is lower than PC.
Profit-maximizing output: MR = MC	Economic cost in MC includes advertising cost for product differentiation.
	PC is efficient as surplus is maximized. PC: Price = Marginal Cost MC: Price > Marginal Cost Deadweight loss in MC because firms have some amount of pricing power and consumer surplus is lost.
	Prices are lower in PC, but consumers have little variety.

5. Oligopoly

Oligopoly and Pricing Strategies

An oligopoly market has few sellers of a product and many buyers. These sellers are large players in their industry who determine the prices or quantities. For example, credit card companies such as Visa, MasterCard, and Amex.

Characteristics**:**

- There are a small number of potential sellers.
- The products offered by each seller are close substitutes for the products offered by the other firms and may be differentiated by brand or be homogeneous and unbranded.
- Entry into the market is difficult with fairly high costs and significant barriers to competition.

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- Firms typically have substantial pricing power. Since there are very few firms, the pricing decisions are interdependent. Whenever a firm makes a decision, it must take into account how the competing firms will react.
- Products are often highly differentiated through marketing, features, and other non-price strategies.
- The pricing is strategic and firms in an oligopoly have a temptation to collude.

Demand Analysis and Pricing Strategies in Oligopoly Markets

If firms collude, the total market demand is divided among the individual participants. The firms act like a cartel and decide how to divide the demand, and what price to set for the products in order to maximize profit.

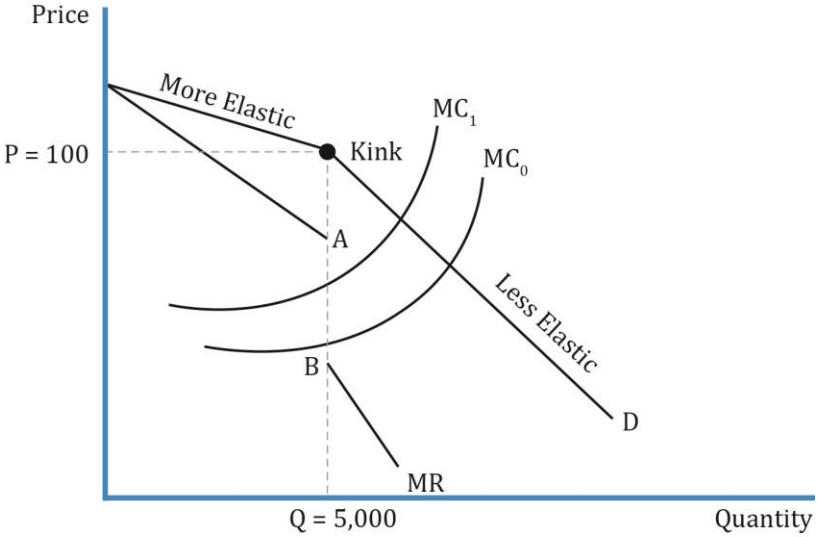
If firms do not collude, each firm faces an individual demand curve and a market demand curve. There are several models that try to explain pricing in oligopoly markets:

- Pricing interdependence
- Cournot assumption
- Nash equilibrium
- Stackelberg model

Pricing Interdependence – Kink Demand Curve

According to this theory, a competitor will not follow a price increase, but will cut prices in response to a price decrease.

Example: Let us assume a town has two cola suppliers: Coke and Pepsi. This type of oligopoly is called a duopoly. Now, assume the initial equilibrium price of 1 liter Coke bottle is 100 and the quantity is 5000.



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Effect of price increase: If Coke increases its price from 100 to 105, what will Pepsi do? According to the interdependence theory, Pepsi will not increase the price and consumers will switch from Coke to Pepsi. The quantity demanded of Coke will decrease (see the elastic portion of the demand curve).

Effect of price decrease: Instead, if Coke decreases the price to 95, then Pepsi will also decrease the price to 95. The quantity demanded of Coke will increase when the price decreases, but not by much because there is no substitution effect. Consumers do not switch from Pepsi to Coke as both are selling at the same price. To the right of the kink, the demand curve is inelastic.

Some important points**:**

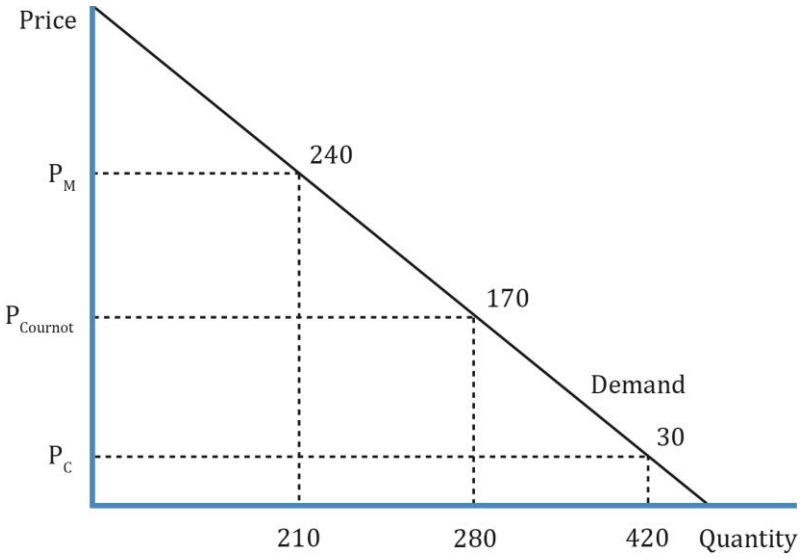
- There are two different demand curves in the model; combining them gives us the overall demand, which is a kinked (bent) curve. A kink in the demand curve leads to a discontinuous (with a gap) marginal revenue curve. One part of the MR curve corresponds to the price increase part of the demand curve and the other to the price decrease part of the demand curve.
- Profit-maximizing rule: MR = MC.
- Equilibrium price and quantity do not change so long as the marginal-cost curve of the firm falls between the gaps in the MR curve.
- The MC must change considerably for the firms to change their price.
- Advantage: The model helps explain stable prices.
- Disadvantage: It does not tell us what the prices should be.

The Cournot Assumption

Firms compete simultaneously to determine a profit-maximizing output, based on the assumption that the other firms' output will not change. In the long run, change in price or quantity will NOT increase profits. As the number of firms in an oligopoly increase, the equilibrium point moves closer to perfect competition.

Assume there are two firms with the output levels q_1 and q_2 respectively. Firm 1 chooses its output as q_1 to maximize profit based on the assumption that firm 2's output level q_2 is constant in the future. Similarly, firm 2 chooses its output as q_2 to maximize profits by assuming that firm 1's output level is constant. Firms choose q_1 and q_2 simultaneously. Let us now look at the price and quantity numbers associated with the Cournot assumption.

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- With a monopoly, the price is highest and quantity is lowest.
- In a perfect competition market structure, the price is lowest and quantity is highest.
- In a duopoly market characterized by the Cournot assumption, both the price and quantity will lie somewhere in between. As the number of firms increase, the equilibrium point moves towards perfect competition.

The Nash Equilibrium

Unlike perfect competition, in oligopoly there is a lot of strategic interdependence between firms. Since the number of firms are few, the actions one takes affects the others.

Nash Equilibrium**:** A set of choices/strategies among two or more participants is called a Nash equilibrium if, holding the strategies of all other participants constant, no participant has an incentive to choose a different strategy. In an oligopoly, firms arrive at an equilibrium strategy after considering the actions of other firms (interdependence). Once they arrive at equilibrium, no firm wants to change its strategy.

Assumptions made in Nash equilibrium**:**

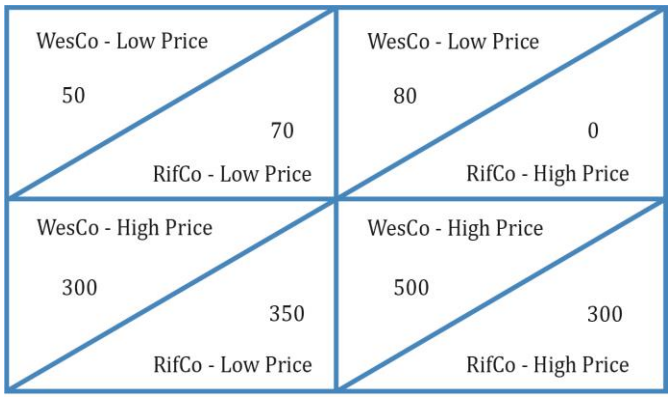
- The firms do the best they can, given the actions of their rivals.
- The actions are interdependent.
- The firms do not cooperate (collude); each firm wants to maximize its own profits.

Example: WesCo and RifCo sell a similar product. Each company can employ a high-price strategy or a low-price strategy. The profit for each strategy is shown. What is the Nash equilibrium?

The four possible strategies are shown in the four boxes. For example, box 1 on the top-left corner has WesCo adopting a low price strategy and RifCo adopting a low price strategy as well. The profit for WesCo is 50 and that for RifCo it is 70. At any point in time, the

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companies can be in only one box. It is not possible for WesCo to adopt a low price strategy with profit of 50 (box 1) and RifCo to adopt a high price strategy with profit of 0 (box 2).



No matter where the companies start, they will end up in box 4 (lower left box).

Let's start with box 1. The total profit of WesCo and RifCo is 120. They are both selling the products at a lower price. It is in WesCo's best interest to increase prices, and their profits jump from 50 to 300 in box 4. It is in RifCo's best interest as well if WesCo increases the price, as RifCo can also increase the price. RifCo's profit jumps from 70 to 350. The combined profit of box 4 now is 650.

The combined profits are the highest in box 3, which is 800. Both the companies are charging high prices. Box 3 is in WesCo's best interest as it earns its maximum profit of 500, but it's possible only if RifCo also charges the high price. But RifCo is not happy here and would lower the prices to increase its profit from 300 (box 3) to 350 (box 4).

When RifCo lowers its price to make a profit of 350 in box 4, WesCo's profit falls from 500 to 300. The Nash equilibrium position in box 4 is what they arrive at finally.

Can both companies be better if they collude? Yes, if both the companies agree to collude and charge high prices. If WesCo and RifCo agree to split the maximum profit of 800 equally, then each company makes a profit of 400, which is better than the Nash equilibrium profit of 300 and 350 profit respectively. Companies are said to form a **cartel** when they engage in collusive agreements openly.

Factors that affect the chances of successful collusion**:**

1. Number and size of sellers: The number of firms should be small.
2. Similarity of products: Products must be homogeneous.
3. Cost structure: Firms must have a similar cost structure.
4. Order size and frequency: Orders must be small and frequent.
5. Retaliation: Less likely to break collusive agreement if the threat of retaliation by other firms is severe.
6. Degree of external competition: Collusion increases overall profitability of the market which attracts external competition.

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Stackelberg Model

There is one dominant large firm and many small firms. The large firm sets the price and has the first mover advantage.

In the Stackelberg model, the decision-making happens sequentially (recall it happens simultaneously in the Cournot assumption). The leader firm chooses the output first and then the follower firm chooses its output.

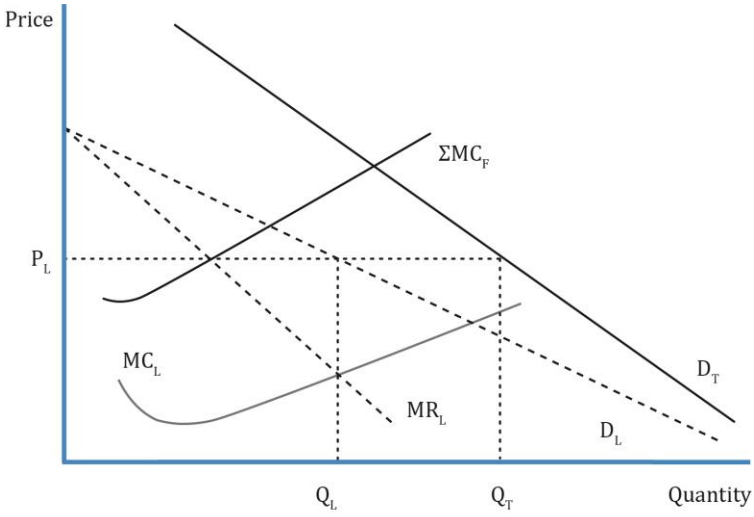
Oligopoly Markets: Optimal Price, Output, and Long-Run Equilibrium

The curriculum discusses the supply analysis for only one type of oligopoly – the dominant firm oligopoly.

- As in monopolistic competition, the supply function is not well defined.
- We cannot determine equilibrium output and price without considering the demand function and competitive strategies.
- Profit-maximizing condition: MR = MC.
- The equilibrium price is based on the demand curve.

Example: Say we have an oligopoly market where one firm has a significantly lower cost of production than its competitors and has a 40% market share. A **dominant** or **leader** firm is a firm with at least 40 % market share, greater capacity, lower cost structure, and is price maker. A follower firm is a small firm that is a price taker – i.e. it accepts the price set by the leader firm. Let us say there are five such firms in this market.

The graph below shows the quantity that will be supplied and the price charged by the market leader, as well as by the other firms.



Interpretation of the graph:

- Price is plotted on the y-axis and quantity on the x-axis.

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- The solid line represents an aggregate market demand. The following are the curves for the dominant firm:
  - Dashed line – the demand curve.
  - MR_L – the marginal revenue curve; it lies below the demand curve and is steeper
  - MC_L – the marginal cost curve.
  - P_L – optimal price.
- Assuming the other five firms will take the price established by the leader, the overall market demand is given as Q_T. The small/follower firms have no incentive to slash prices as it will lead to price wars with the leader, who is a low-cost producer.
- Quantity supplied by the leader firm = Q_L; quantity supplied by the other firms = $Q_T - Q_L$.
- Notice that the demand curves of the industry and of the dominant firm are not parallel to each other. As the price decreases, the difference between the curves diminishes. The reasons are:
  - The dominant firm is a low cost producer. When prices start falling, the other smaller firms exit the industry because they do not want to sell below cost.
  - The dominant firm gets a greater market share as other firms exit, and Q_L (quantity supplied by the leader) increases.

Optimal Price and Output

There is no single optimum price and output model that works for all oligopoly market situations because of different strategies and pricing methods. The process for determining the optimal price for a few methods is listed below:

- Kinked demand curve: Price at the kink in demand function.
- Dominant firm: Price at the quantity where MR = MC. Followers take the leader's price.
- Cournot assumption: No changes in price and output by other firms once the dominant firm chooses its output level where MR=MC.
- Nash equilibrium: Each firm acts in its best interest under the given circumstances. No certainty of price and output level.

Factors Affecting Long-Run Equilibrium in Oligopoly Markets

Long-run economic profits are possible, but empirical evidence suggests that over time the market share of the dominant firm declines.

6. Determining Market Structure

Analysts are interested in investing in markets with high pricing-power as it drives profitability. If there are very few large firms in an industry, then the price tends to be high and the quantity supplied low. When there is a possible merger, analysts should consider the impact of competition law (anti-trust law) as regulators might prevent the merger to keep

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the industry competitive. In many countries, competition law has been introduced to regulate the degree of market competition in different industries of different countries.

Econometric Approaches

Econometric approaches can be used for measuring market concentration or market power. Some key points in this context are as follows:

- Use regression analysis to estimate elasticity of demand and supply.
- If demand is inelastic, then it indicates companies may have market power.
- The disadvantage is that though it is theoretically appealing, but data is not easily available.

Simpler Measures

Simpler approaches to estimate elasticity that avoid the drawbacks in regression analysis include the N-firm concentration ratio and Herfindahl-Hirshman Index (HHI).

N-Firm Concentration Ratio and HHI

N-Firm concentration ratio: It is the sum of the market shares of the largest N firms. It is almost zero for perfect competition and 100 for monopoly.

For example, in an industry, assume the five largest firms in the industry have a market share of 25%, 15%, 10%, 10% and 10%. The 5-firm concentration ratio would be 70%.

Advantages:

- Data is easily available.
- Simple to use and understand.

Disadvantages**:**

- Unaffected by mergers among top firms. Assume the top two firms by market share merge and the market shares of five largest firms are 40%, 10%, 10%, 10% and 2% now. The 5-firm concentration ratio would be 72% instead of 70%, which is not very different from what it was earlier. But the largest firm has a high market share of 40%, which is not evident in the concentration ratio number.
- Does not quantify market power.
- Does not consider barriers to entry.
- Does not consider elasticity of demand.

Herfindahl-Hirschman Index (HHI)

- Sum of squared market shares of N largest firms in a market (ranges from 0 to 1). A number close to 1 indicates it is concentrated or monopolistic.
- For example, assume the market shares of four firms are 50%, 20%, 10% and 20%. The HHI is $0.5^2 + 0.2^2 + 0.1^2 + 0.2^2 = 0.34$.

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Advantage**:**

Simple and commonly used by regulators.

Disadvantage**:**

- Does not consider barriers to entry.
- Does not consider elasticity of demand.

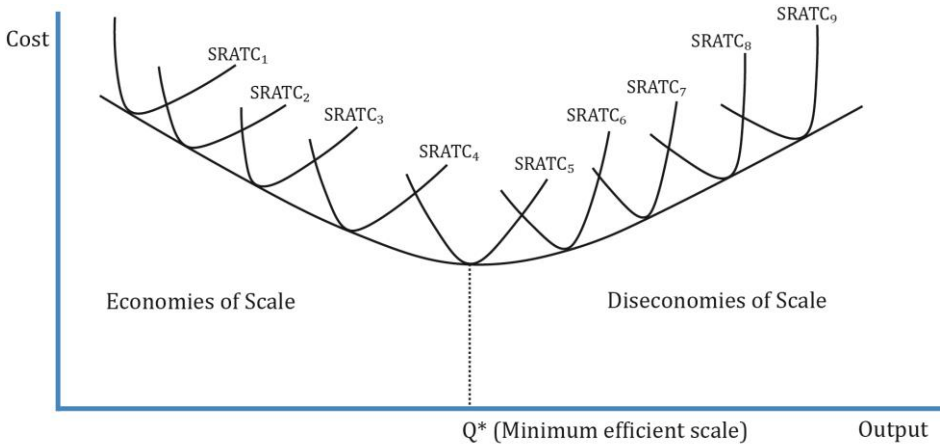
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Summary

LO: Determine and interpret breakeven and shutdown points of production, as well as how economies and diseconomies of scale affect costs under perfect and imperfect competition.

Situation	Short Run	Long Run
Price > ATC	Economic Profit - Operate	Economic Profit - Operate
Price = ATC	Breakeven - Operate	Breakeven - Operate
AVC < Price < ATC	Operate	Shutdown
Price < AVC	Shutdown	Shutdown

Economies of scale: As output increases the long-run cost per unit decreases.
Diseconomies of scale: As output increases the long-run cost per unit increases.



LO: Describe characteristics of perfect competition, monopolistic competition, oligopoly, and pure monopoly.

	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of Sellers	Many firms	Many firms	Few firms	Single firm
Barriers to Entry	Very low	Low	High	Very high
Nature of Substitute Products	Very close substitutes	Substitutes but differentiated	Very close substitutes or differentiated	No good substitutes
Nature of Competition	Price only	Price, marketing, & features	Price, marketing, & features	Advertising
Price Power	None	Some	Some to significant	Significant

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LO: Explain supply and demand relationships under monopolistic competition, including the optimal price and output for firms as well as pricing strategy.

- Since the products are unique, a monopolistic firm has a downward sloping demand curve. MR is steeper and lies below the demand curve.
- Profit maximization output is based on MR = MC.
- Price is determined based on the demand curve.
- The supply function is not well-defined in monopolistic competition.

LO: Explain supply and demand relationships under oligopoly, including the optimal price and output for firms as well as pricing strategy.

If firms do not collude, each firm faces an individual demand curve and a market demand curve. There are several models that try to explain pricing in oligopoly markets:

- Pricing interdependence
- Cournot assumption
- Nash equilibrium
- Stackelberg model

The curriculum discusses the supply analysis for only one type of oligopoly – the dominant firm oligopoly.

- As in monopolistic competition, the supply function is not well defined.
- We cannot determine equilibrium output and price without considering the demand function and competitive strategies.
- Profit-maximizing condition: MR = MC.
- The equilibrium price is based on the demand curve

There is no single optimum price and output model that works for all oligopoly market situations because of different strategies and pricing methods. The process for determining the optimal price for a few methods is listed below:

- Kinked demand curve: Price at the kink in demand function.
- Dominant firm: Price at the quantity where MR = MC. Followers take the leader's price.
- Cournot assumption: No changes in price and output by other firms once the dominant firm chooses its output level where MR=MC.
- Nash equilibrium: Each firm acts in its best interest under the given circumstances. No certainty of price and output level.

LO: Identify the type of market structure within which a firm operates and describe the use and limitations of concentration measures.

N-Firm Concentration Ratio:

- Sum of the percentage market shares of the N largest firms in an industry.
- Market share = Firm sales / Total market sales.

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- Advantage: Simple to use and understand.
- Disadvantages: Ignores barriers to entry and does not directly measure market power or elasticity of demand.

Herfindahl-Hirschman Index (HHI):

- HHI = sum of squared market shares of N largest firms in a market
- Ranges from 0 to 1 – where 0 indicates perfect competition and 1 indicates a perfect monopoly.
- Advantage: Simple and commonly used by regulators.
- Disadvantage: Does not consider barriers to entry and elasticity of demand.

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LM02 Understanding Business Cycles

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Version 1.0

1. Introduction

This learning module covers:

- What is a business cycle; what are the different phases in a business cycle?
- Credit cycles and their relationship to business cycles
- How resource use, capital spending, and inventory levels vary over the business cycle
- Economic indicators that are useful in predicting the future of an economy.

2. Overview of the Business Cycle

The curriculum defines business cycle as: "Business cycles are a type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in **business enterprises**: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle; this sequence of events is **recurrent but not periodic**; in duration, business cycles vary from more than one year to 10 or 12 years."

Some important points to be noted from the definition:

- Business cycles occur in economies where there are a large number of private companies, and not just agriculture economies.
- The economic activity shows a cyclical behavior between expansion and recession.
- They are pervasive, i.e., the cycle includes many economic activities and not just one sector. And the phases of expansion or contraction occur at the same time throughout the economy. For example, banking and real estate both may be in an expansion stage.
- They are recurrent but not periodic, i.e., the cycles repeat. To say they are not periodic means that the intensity and the duration differs. For instance, if an economic boom lasted for five years from 2002-07, it does not mean that the expansion phase will last for five years in the next cycle. Each cycle lasts about 1 to 12 years.

Phases of the Business Cycle

Types of Cycles

'Classical cycle' refers to fluctuations in the level of economic activity when measured by GDP in volume terms. It is rarely used in practice, because it does not easily allow us to distinguish between short-term fluctuations and long-term trends.

'Growth cycle' refers to fluctuations in economic activity around the long-term potential trend growth level. The focus is on how much actual economic activity is below or above trend growth in economic activity. The advantage of this method is that it divides economic activity into a part reflecting long-term trends and a part reflecting short-term fluctuations.

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'Growth rate cycle' refers to fluctuations in the growth rate of economic activity. A growth rate above potential growth rate reflects upswings, while a growth rate below potential growth rate reflects downswings. The advantage of this definition is that there is no need to estimate a long-run growth path first. Also, peaks and troughs can be recognized earlier than when using the other two definitions.

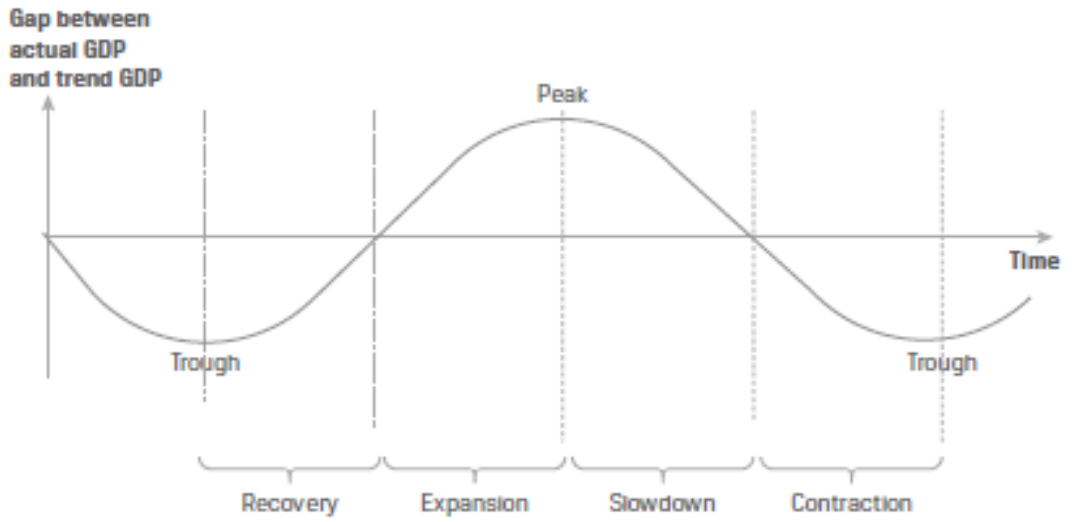
Four Phases of the Cycle

There are four phases of a business cycle:

1. Recovery: The economy is going through the trough of the cycle. Economic activity (which includes consumer and business spending) is below potential but is starting to increase.
2. Expansion: The recovery gathers momentum and economic activity rises above potential. The economy enters the so-called 'boom' phase.
3. Slowdown: The economy is going through the peak of the cycle. Economic activity is above potential but is starting to decrease.
4. Contraction: Economic activity falls below potential. The economy may experience a recession.

The four stages are illustrated below (this exhibit is reproduced from the curriculum):

Schematic of Business Cycle Phases



Some of the important characteristics of each phase in a business cycle are presented in Exhibit 5 from the curriculum.

Characteristics:

Phase	Recovery	Expansion	Slowdown	Contraction
Description	Economy going through a trough.	Economy enjoying an upswing.	Economy going through a peak. Positive	Economy weakens and may go into a recession.

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	Negative output gap starts to narrow.	Positive output gap opens.	output gap starts to narrow.	Negative output gap opens.
Activity levels – consumers and businesses	Activity levels are below potential but start to increase.	Activity measures show above-average growth rates.	Activity measures are above average but decelerating. Moving to below-average rates of growth.	Activity measures are below potential. Growth is lower than normal.
Employment	Layoffs slow. Businesses rely on overtime before moving to hiring. Unemployment remains higher than average.	Businesses move from using overtime and temporary employees to hiring. Unemployment rate stabilizes and starts falling.	Businesses continue hiring but at a slower pace. Unemployment rate continues to fall but at decreasing rates.	Businesses first cut hours, eliminate overtime, and freeze hiring, followed by outright layoffs. Unemployment rate starts to rise.
Inflation	Inflation remains moderate.	Inflation picks up modestly	Inflation further accelerates.	Inflation decelerates but with a lag.

Leads and Lags in Business and Consumer Decision Making

The behavior of businesses and households often leads or lags the turning points of the business cycle. For example, at the start of an expansion phase, businesses rely on overtime and wait to hire new employees until they are certain that the economy is truly growing.

Market Conditions and Investor Behavior

Recovery phase: When markets expect the end of a recession and the start of an expansion phase, risky assets are repriced upwards. Investors start incorporating higher profit expectations into the prices of stocks and bonds. Equity markets typically hit a trough about 3-6 months before the economy hits the trough.

Expansion phase: The boom phase tests the limits of the economy. For example, companies may expand so much that they have difficulty finding qualified labor and will compete with other companies by raising wages. Companies may also borrow a lot to fund capacity expansions. The government may have to step in to prevent the economy from overheating.

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Slowdown phase: During the boom, the riskiest assets will usually have significant price increases. Whereas, safe assets such as government bonds may have lower prices and thus higher yields.

Contraction phase: During contraction, investors prefer safer assets such as government securities and defensive companies with stable positive cash flows.

3. Credit Cycles

Credit cycles describe the changing availability—and pricing—of credit. When the economy is strong the willingness of lenders to extend credit on favorable terms is high. Whereas, when the economy is weak lenders make credit less available and more expensive. This can result in decline of asset values and cause further economic weakness and higher defaults.

Applications of Credit Cycles

Credit cycles should be studied due to the importance of credit in the financing of construction and the purchase of property. The duration of recessions and recoveries are often shaped by linkages between business and credit cycles. Recessions accompanied by rapid fall in credit tend to be longer and deeper. Such situations can also lead to housing and equity price busts. Recoveries accompanied by rapid growth in credit tend to be stronger. They can also lead to a revival in house and equity prices.

Credit cycles are not always synchronized with the business cycle. They tend to be longer, deeper, and sharper than the business cycle.

Consequences for Policy

Investors pay attention to the stage in the credit cycle because:

- it helps them understand developments in the housing and construction markets
- it helps them assess the extent of business cycle phases
- it helps them better anticipate policy makers' actions

4. Economic Indicators Over the Business Cycle

The Workforce and Company Costs

Exhibit 7 from the curriculum depicts the pattern of hiring and employment through different phases of the business cycle.

Phase	Recovery	Expansion	Slowdown	Contraction
Description of activity levels	Economy starts at trough and output below potential. Activity picks up, and gap starts to close.	Economy enjoying an upswing, with activity measures showing above-average growth rates.	Economy at peak. Activity above average but decelerating. The economy may experience shortages of factors of production as	Economy goes into a contraction, (recession, if severe). Activity measures are below potential. Growth is lower than normal.

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			demand may exceed supply.	
Employment	Layoffs slow. Businesses rely on overtime before moving to hiring. Unemployment remains higher than average.	Businesses move from using overtime and temporary employees to hiring. Unemployment rate stabilizes and starts falling.	Businesses continue hiring but at a slower pace. Unemployment rate continues to fall but at slowly decreasing rates.	Businesses first cut hours, eliminate overtime, and freeze hiring, followed by outright layoffs. Unemployment rate starts to rise.
	Levels of employment lag the cycle			

Fluctuations in Capital Spending

Exhibit 8 from the curriculum depicts the fluctuations in capital spending through different phases of the business cycle.

Phase	**Recovery**	**Expansion**	**Slowdown**	**Contraction**
Business conditions and expectations	Excess capacity during trough, low utilization, little need for capacity expansion. Interest rates tend to be low—supporting investment.	Companies enjoy favorable conditions. Capacity utilization increases from low levels. Over time, productive capacity may begin to limit ability to respond to demand. Growth in earnings and cash flow gives businesses the financial ability to increase investment spending.	Business conditions at peak, with healthy cash flow. Interest rates tend to be higher—aimed at reducing overheating and encouraging investment slowdown.	Companies experience fall in demand, profits, and cash flows.
Capital spending	Low but increasing as companies start to enjoy better conditions. Capex focus on efficiency rather than capacity. Upturn most pronounced in orders for light producer equipment. Typically, the orders initially reinstated are for equipment with a high rate of obsolescence, such as software,	Customer orders and capacity utilization increase. Companies start to focus on capacity expansion. The composition of the economy's capacity may not be optimal for the current structure of demand, necessitating spending on new types of equipment. Orders precede actual shipments, so orders for capital equipment are a widely watched	New orders intended to increase capacity may be an early indicator of the late stage of the expansion phase. Companies continue to place new orders as they operate at or near capacity.	New orders halted, and some existing orders canceled (no need to expand). Initial cutbacks may be sharp and exaggerate the economy's downturn. As the general cyclical bust matures, cutbacks in spending on heavy equipment further intensify the contraction. Maintenance scaled back.

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	systems, and technological hardware.	indicator of the future direction of capital spending.		
Examples	Software, systems, and hardware (high rates of obsolescence) orders placed or re-instated first.	Heavy and complex equipment, warehouses, and factories. A company may need warehouse space in locations different from where existing facilities are.	Fiber-optic overinvestment in late 1990s that peaked with the "technology, media, telecoms bubble."	Technology and light equipment with short lead times get cut first. Cuts in construction and heavy equipment follow.

Fluctuations in Inventory Levels

Increase and decrease in inventory happens very rapidly, and has a major effect on economic growth despite the small size. $\frac{\text{Inventory}}{\text{Sales}}$ (I/S) ratio is an important indicator. Final sales numbers better reveal the reality of the economic situation than inventory numbers because the inventory may accumulate or companies may want to dispose obsolete inventory before starting production; it depends on the stage of the business cycle. Inventories tend to rise when the I/S ratio is low. During recovery, inventory will be less than sales and companies start production to increase inventory.

Exhibit 9 from the curriculum shows how production, sales, and inventories typically move through different phases of the business cycle.

Phase	**Recovery**	**Expansion**	**Slowdown**	**Contraction**
Sales and production	Sales decline slows. Sales subsequently recover. Production upturn follows but lags behind sales growth. Over time, production approaches normal levels as excess inventories from the downturn are cleared.	Sales increase. Production rises fast to keep up with sales growth and to replenish inventories of finished products. This increases the demand for intermediate products. "Inventory rebuilding or restocking stage."	Sales slow faster than production; inventories increase. Economic slowdown leads to production cutbacks and order cancellations.	Businesses produce at rates below the sales volumes necessary to dispose of unwanted inventories.
Inventory–sales ratio	Begins to fall as sales recovery outpaces production.	Ratio stable.	Ratio increases. Signals weakening economy.	Ratio begins to fall back to normal.

Economic Indicators

Economic indicators are variables that are used to assess the state of the overall economy and for providing insights into future economic activity.

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Types of Indicators

Economic indicators can be classified based on whether they lag, lead, or coincide with changes in an economy's growth.

- *Leading indicators* have turning points that tend to precede those of the business cycle. They help in forecasting the economy in the near term.
  - Ex: Weekly hours in manufacturing, S&P 500 return, private building permits.
- *Coincident indicators* have turning points that tend to coincide with those of the business cycle and are used to indicate the current phase of the business cycle.
  - Ex: Manufacturing activity, personal income, number of non-agricultural employees.
- *Lagging indicators* have turning points that tend to occur after those of the business cycle.
  - Ex: Bank prime lending rate, inventory-to-sales ratio, average duration of unemployment.

Composite Indicators

Composite indicators consist of a composite of different variables that all tend to move together. Different countries will have different composite indices. These indicators are based on empirical observations of an economy.

Leading Indicators

In the U.S., the composite leading indicator is called the Index of Leading Economic Indicators (LEI) that consists of 10 component parts. Exhibit 11 presents the 10 components used in the LEI.

1. Average weekly hours, manufacturing: Firms cut on overtime before a downturn and increase the overtime before hiring full-time workers during a recovery.
2. Average weekly initial claims for unemployment insurance.
3. Manufacturers' new orders for consumer goods and materials.
4. ISM new order index: The Institute of Supply Management (ISM)polls its members to build indexes of manufacturing orders, output, employment, pricing, and comparable gauges for services.
5. Manufacturers' new orders for non-defense capital goods excluding aircraft
6. Building permits for new private housing units.
7. S&P 500 stock index.
8. Leading credit index: Aggregates the information from six leading financial indicators, which reflect the strength of the financial system to endure stress.
9. Interest rate spread between 10-year Treasury yields and overnight borrowing rates: Spread is the difference between long-term yields and short-term yields. If the curve is upward sloping (a wider spread), then we expect short-term rates in the future to be high and more economic growth.

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10. Average consumer expectations for business conditions

Using Economic Indicators

Leading indicators are particularly useful as they can help predict where the economy is likely to be in the near future. Coincident and lagging indicators can help confirm what the leading indicators are telling us.

Other Composite Leading Indicators

Apart from the LEI used in the US, there are several other composite leading indicators that are used across different countries.

Surveys

Composite indicators often make use of economic surveys. These surveys are usually conducted by central banks, research institutes, statistical offices, and trade associations on a monthly or quarterly basis. The surveys are conducted among either businesses, consumers, or experts. For example, among the 10 components of the LEI, the following three components are survey based:

- ISM new order index
- Manufacturers' new orders for non-defense capital goods excluding aircraft
- Average consumer expectations for business conditions

The Use of Big Data in Economic Indicators

In recent years, due to the vast increase in the amount of information and the number of variables that go into composite indicators, statistical techniques such as 'principal component analysis' are frequently used while constructing indexes using economic indicators.

Nowcasting

Policy makers and market practitioners have started monitoring economic and financial variables such as internet searches and electronic payment data in real-time. This allows them to continuously assess current conditions and produce a nowcast.

Nowcasting produces an estimate of the current state of the economy. The advantage of this method is that it helps overcome delays associated with the release of actual measures. For example, measures such as GDP are only published with delay, after the end of the time period under consideration.

GDPNow

GDPNow is a nowcast published by the Federal Reserve Bank of Atlanta. According to the Atlanta Fed, "GDPNow" is "best viewed as a running estimate of real GDP growth based on available data for the current measured quarter."

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Summary

LO: Describe the business cycle and its phases.

There are four phases of a business cycle:

1. Recovery: The economy is going through the trough of the cycle. Economic activity (which includes consumer and business spending) is below potential but is starting to increase.
2. Expansion: The recovery gathers momentum and economic activity rises above potential. The economy enters the so-called 'boom' phase.
3. Slowdown: The economy is going through the peak of the cycle. Economic activity is above potential but is starting to decrease.
4. Contraction: Economic activity falls below potential. The economy may experience a recession.

LO: Describe credit cycles.

Credit cycles describe the changing availability—and pricing—of credit. When the economy is strong the willingness of lenders to extend credit on favorable terms is high. Whereas, when the economy is weak lenders make credit less available and more expensive. This can result in decline of asset values and cause further economic weakness and higher defaults.

LO: Describe how resource use, consumer and business activity, housing sector activity, and external trade sector activity vary over the business cycle and describe their measurement using economic indicators.

Recovery:

- GDP growth rate changes from negative to positive.
- High unemployment rate and a moderate or declining inflation.
- Increasing production to meet the pickup in sales with more flexible methods like overtime or increasing utilization levels.

Expansion:

- GDP growth rate increases.
- Reduction in unemployment rate as hiring rises.
- Inflation may begin to rise.
- Increasing production needs are met with investments and labor force additions.

Slowdown:

- GDP growth rate decreases.
- Unemployment rate decreases, but firms cut back on hiring.
- Business and consumer confidence declines, slowing the growth rates in investments and consumer spending.
- Inflation rate increases.

Contraction:

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- GDP growth rate is declining.
- Unemployment rate increases as firms cut back on production.
- Inflation decreases with a lag.
- Decline in consumer and business confidence lowers the investment and consumer spending.

The three types of indicators are:

- Leading indicators:
  - Indicators that change direction before the peaks and troughs of business cycles.
  - Examples: Stock prices, index of consumer expectations etc.
- Coincident indicators:
  - Indicators that change direction at roughly the same time as the peaks and troughs of business cycles.
  - Examples: Industrial production, manufacturing and trade sales etc.
- Lagging indicators:
  - Indicators that do not change direction until the expansions and contractions are already in progress.
  - Examples: inventory-sales ratio, unemployment rate, labor cost per unit of output, etc.

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LM03 Fiscal Policy

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Ver 1.0

1. Introduction

This learning module covers:

- An introduction to monetary and fiscal policy
- Roles and objectives of fiscal policy
- Tools of fiscal policy and their advantages and disadvantages
- Fiscal policy implementation and difficulties faced in implementation

2. Introduction to Monetary and Fiscal Policy

As compared to households and corporations, the economic decisions made by governments can have an enormous impact on economies because governments are usually the largest employers, largest spenders and largest borrowers in an economy.

There are two types of government policy:

Fiscal policy**:** Refers to government decisions about taxation and spending.

Monetary policy**:** Refers to central bank activities directed towards influencing the quantity of money and credit in the economy.

The overall goal of these policies is to create an economic environment in which growth is stable and positive and inflation is stable and low.

3. Roles and Objectives of Fiscal Policy

Fiscal policy refers to the taxing and spending policies of the government. A government can influence the following aspects of the economy:

- Overall level of aggregate demand in an economy and hence the level of economic activity. (This is often the primary objective of a fiscal policy; secondary objectives, which are tied to the political motive of the government, are as follows:)
- Distribution of income and wealth among different segments of the population.
- Allocation of resources between different sectors and economic agents.

Roles and Objectives of Fiscal Policy

Fiscal policy can be contractionary or expansionary.

An expansionary fiscal policy can take several forms:

- Lower taxes
  - Cuts in personal income tax (This increases the disposable income).
  - Cuts in sales taxes (This lowers the prices).
  - Cuts in corporate taxes increase business profits (This means that corporates have more money to invest).
- Higher government spending on social goods and infrastructure.

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Contractionary fiscal policy**:** It is the opposite of expansionary fiscal policy. Higher taxes or lower government spending are examples of contractionary fiscal policy.

What are the Keynesian and Monetarist views on the effectiveness of fiscal policy?

Keynesian view**:** Government intervention is necessary in the form of fiscal policy to get an economy out of recession. They believe that the aggregate demand, employment, and output increase with fiscal policy.

Monetarist view**:** Monetary policy is a more effective tool to tame inflation; monetarists advocate a steady, stable monetary policy. They believe that Fiscal policy only has a temporary effect.

Government Receipts and Expenditures in Major Economies

- Exhibits 2 and 3 from the curriculum show government revenues and expenditures as a percentage of GDP for some of the major economies. As of 2015, for the U.S., government revenue as a percent of GDP was 33.4%, while the government expenditure as a percent of GDP was 37.6%.
- The possibility that fiscal policy can influence output means that it can be used to stabilize an economy.
- The budget deficit is the difference between government revenue and expenditure for a fixed period of time. Government revenue includes tax revenues, net of transfer payments; government spending includes interest paid on government debt.
- An increase in budget surplus indicates a contractionary fiscal policy.
- An increase in budget deficit indicates an expansionary fiscal policy.
- Two fiscal policies to stabilize the economy include:
  - Automatic stabilizers: When the economy slows and unemployment rises, government spending on social insurance and unemployment benefits will rise. Whereas, if the economy is at full employment, taxes collected will be high and there will be a budget surplus. These happen automatically without the intervention of policymakers, and the focus is primarily on aggregate demand. They help reduce the impact of a recession.
  - Discretionary fiscal policies: Changes in government spending or tax rates. In contrast to automatic stabilizers, this depends on the policy makers. The policies differ primarily with respect to timing.
- A balanced budget is one where government spending is equal to government revenues.

Deficits and the National Debt

Government deficit = Revenue – Expenses

Government deficit (national debt) is the accumulation of these deficits over time. Should we worry about national debt? There are two schools of thought.

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We should not worry because:

- The scale of the problem may be overstated because the debt is owed internally to fellow citizens.
- A proportion of the money borrowed may have been used for capital investment projects or enhancing human capital. We are borrowing now to increase our productive capacity in the future.
- Large fiscal deficits require tax changes that may actually reduce distortions caused by existing tax structures.
- Deficits may have no net impact because the private sector may act to offset fiscal deficit by increasing saving in anticipation of future increased taxes. This is known as Ricardian equivalence**.**
    - The government funds its spending by either increasing taxes or borrowing. It is the future taxpayers who will service the government's debt. So, it is the taxpayers who bear the burden in both cases. What matters is only the timing: now or later. According to Ricardian equivalence, if the government defers taxation, consumers anticipate higher taxes and the private sector will save enough today to pay for increased taxes in the future. This higher saving results in decreased private demand and increased government demand. The net effect is offsetting, as government spending does not create demand stimulus.
- If there is unemployment in an economy, then the debt is not diverting activity away from productive uses.

We should worry because:

- High levels of debt to GDP may lead to higher tax rates in the search for higher tax revenues. This could create disincentives for economic activity.
- If markets lose confidence in a government, then the central bank may have to print money to finance a government deficit. This may lead to inflation.
- Government borrowing may divert private sector investment from taking place (this effect is called crowding out). If savings are limited and the demand for funds from the government is high, then it will lead to higher interest rates and lower private sector investment.

4. Fiscal Policy Tools

- Government spending can take different forms:
    - Transfer payments**:** Welfare payments provided to low-income households so that they get a basic minimum level of income. Not included in GDP calculation. Ex: pensions, housing, and unemployment benefit, etc.
    - Current government spending**:** Regular spending on goods and services such as education, healthcare, defense, etc.

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  - Capital expenditure**:** Spending on infrastructure such as building roads, schools, hospitals, etc.

- Government spending is justified both on economic and social grounds as they ensure employment, economic growth, and a minimum standard of living for lower income households.
- Government revenue can take different forms.
  - Direct taxes: Taxes imposed on income, property, wealth, corporate profits, capital gains, inheritance, etc. These include taxes levied on individuals and businesses.
  - Indirect taxes: Taxes imposed on goods and services such as excise duty, VAT. Indirect taxes affect alcohol or tobacco consumption more directly than direct taxes.
- Following are the desirable attributes of tax policy:
  - Simplicity**:** There should be no ambiguity, loopholes, or scope of interpreting the tax liability differently. It should be simple for the taxpayer to adhere to the rules, and the authority to enforce.
  - Efficiency**:** The tax policy should interfere as little as possible in the choices individuals make in the market place.
  - Fairness**:** Are people in similar situations levied the same tax, or are rich people taxed more? For example, should a person earning $1 million a year be in the same tax bracket as one earning $50,000 a year? It is subjective.
  - Revenue sufficiency**:** Tax revenues collected should be sufficient to cover expenditure.

The Advantages and Disadvantages of Different Fiscal Policy Tools

Advantages and disadvantages of using different tools of fiscal policy	
Advantages	**Disadvantages**
Indirect taxes (such as VAT) can be adjusted almost immediately after they are announced and can influence spending behavior instantly. Generates revenue for the government at little or no cost to the government.	Direct taxes are more difficult to change without considerable notice, often many months, because payroll computer systems will have to be adjusted. For instance, the government cannot increase income tax every year.
Social policies such as discouraging alcohol or use of tobacco can be adjusted almost instantly by raising such taxes.	The same may be said for welfare and other social transfers.
	Capital spending plans (building highways or schools) take longer to formulate and implement; typically, over a period of years.

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Modeling the Impact of Taxes and Government Spending: The Fiscal Multiplier

- The objective of fiscal policy is to influence output through changes in government spending and/or taxes.
- The fiscal multiplier tells us about changes in output when there are changes in spending and taxes.
- The derivation for the fiscal multiplier is given in the curriculum, but from an exam perspective, it is important to know the formula given below:

  $$\text{Fiscal multiplier} = \frac{1}{1 - c(1 - t)}$$

  where:
  c = marginal propensity to consume
  t = tax rate

  The fiscal multiplier is inversely related to the tax rate and directly related to the marginal propensity to consume.

For example, what is the value of the fiscal multiplier if the tax rate is 20%, and the marginal propensity to spend is 90%?

What is the increase in total income if government spending increases by $1 billion?

Solution:

$$\text{Fiscal multiplier} = \frac{1}{1 - 0.9\,(0.8)} = 3.57$$

A $1 billion increase in government spending increases total income by $3.57 billion.

The Balanced Budget Multiplier

A balanced budget is a fiscal policy tool where the increase in government spending on goods and services is equal to the increase in tax revenues. The net effect is that there is no change in the budget deficit or surplus.

Since it is a balanced budget, government expenditure and taxes go up by the same amount. If this is the case, then the aggregate output actually rises. How? Because the fiscal multiplier is a function of marginal propensity to consume, c. Since c is less than 1, output Y increases. We will see how this happens using an example.

Assume in equilibrium, output Y = 1,000; C = 900 and I = 100. Assume government spending increases by 200, which is financed by an increase in tax revenue of 200. MPC = 0.9

Fiscal multiplier effect = 10

Taxes increase by 200. Disposable income decreases by 200.

Consumption decreases by 0.9 * 200 = 180

Initial impact on aggregate demand = 200 - 180 = 20

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Impact on output because of multiplier effect = 20 * 10 = 200

5. Fiscal Policy Implementation

- The deficit may not be an indication of the government's fiscal stance because an economy goes through a cycle. For example, at the peak of a cycle, unemployment would be low and government expenditure would be less with the likelihood of running a surplus. Similarly, if the economy is in a recession year, then incomes are low and taxes collected will be relatively low, causing the budget deficit to increase. So, one cannot conclude if the government is following a contractionary or expansionary policy by simply looking at the deficit.
- To get an idea of the government's policy, one should look at the structural or cyclically adjusted budget deficit. This is the deficit if the economy was at full employment. If the output is at long-run equilibrium, then the surplus or deficit would be called the structural or cyclically adjusted budget deficit.
- Automatic stabilizers such as social security payments, progressive income taxes, and VAT must be considered to determine the fiscal stance. As unemployment rises, the benefits increase and net tax revenues decrease. These do not require policy changes, and automatically kick in to stimulate growth.
- In addition, there are also discretionary fiscal adjustments used by governments, such as tax changes, or huge spending to build a highway system in a country, to increase aggregate demand.
- The two approaches to fiscal policy vary primarily with respect to timing of implementation. But fiscal policy does not always stabilize an economy, as executing fiscal policy can be difficult for the following reasons:
    - Recognition lag: There is a time lag before policymakers recognize whether the economy is going through a boom or is in recession. This is because it takes time to gather and collate the data: indicators such as unemployment and inflation are often presented weeks later. This problem is generally referred to as driving by looking in the rear-view mirror.
    - Action lag: Once the policymakers acknowledge the problem (recession, or economy slowing down, or inflation), then they have to decide on an action plan. The appropriate policy takes time to implement and must be passed through the congress/parliament/whatever is appropriate. For instance, increased spending on infrastructure to generate employment and boost growth may take several months to complete.
    - Impact lag: It may be a while before the result of the projects undertaken can be seen.
- The timing of the policy action is critical. It is important to understand the course of the economy without these policy changes. Is the economy in recovery mode because of a

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surprise increase in investment spending? Some issues associated with discretionary fiscal adjustments are:

- If a government is concerned with unemployment and inflation, then increasing AD to full employment may push prices further up.
- If the deficit is already large relative to GDP, then it may be difficult for the government to borrow more money to provide fiscal stimulus. Interest on government debt would rise.
- Crowding out effect: Limited savings and increase in government spending → investment available for private sector decreases → less investment spending → less growth.

- Macroeconomic forecasting models are not accurate and cannot be used for policymaking decision effectively.

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Summary

LO: Compare monetary and fiscal policy.

Monetary policy refers to central bank activities directed towards influencing the quantity of money and credit in the economy.

Fiscal policy refers to government decisions about taxation and spending.

LO: Describe roles and objectives of fiscal policy as well as arguments as to whether the size of a national debt relative to GDP matters.

Primary objective is to manage the economy through spending and taxation and their influences on aggregate national output (real GDP). Secondary objectives include:

- Distribution of income and wealth among different segments of the population.
- Allocation of resources between different sectors and economic agents.

The fiscal multiplier tells us about changes in output when there are changes in spending and taxes. Higher future taxes lead to disincentives to work, negatively affecting long-term economic growth. Fiscal deficits may not be financed by the market when debt levels are high.

A crowding-out effect may occur as government borrowing increases demand for loanable funds, thus leading to increased interest rates and decreased private sector investments. Deficits for capital spending can boost the productive capacity of the economy.

Ricardian equivalence may prevail: private savings rise in anticipation of the need to repay principal on government debt. At below full employment levels, deficits do not necessarily crowd out private investment.

LO: Describe tools of fiscal policy, including their advantages and disadvantages.

Government spending can take different forms such as transfer payments, current government spending, and capital expenditure.

Government revenue can be in the form of direct and indirect taxes.

Advantages and disadvantages of using different tools of fiscal policy	
Advantages	**Disadvantages**
Indirect taxes (such as VAT) can be adjusted almost immediately after they are announced and can influence spending behavior instantly. Generates revenue for the government at little or no cost to the government.	Direct taxes are more difficult to change without considerable notice, often many months, because payroll computer systems will have to be adjusted. For instance, the government cannot increase income tax every year.

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Social policies such as discouraging alcohol or use of tobacco can be adjusted almost instantly by raising such taxes.	The same may be said for welfare and other social transfers.
	Capital spending plans (building highways or schools) take longer to formulate and implement, typically over a period of years.

LO: Explain the implementation of fiscal policy and difficulties of implementation as well as whether a fiscal policy is expansionary or contractionary.

Fiscal policy is difficult to execute because it suffers from the following lags:

- Recognition lag: Time taken to recognize that policy needs changes.
- Action lag: Time taken by governments to enact required fiscal policy changes.
- Impact lag: Time taken for fiscal policy to affect economic activity.

Fiscal policy can be contractionary or expansionary.

An expansionary fiscal policy can take several forms:

- Lower taxes
  - Cuts in personal income tax (This increases the disposable income).
  - Cuts in sales taxes (This lowers the prices).
  - Cuts in corporate taxes increase business profits (This means that corporates have more money to invest).
- Higher government spending on social goods and infrastructure.

Contractionary fiscal policy**:** It is the opposite of expansionary fiscal policy. Higher taxes or lower government spending are examples of contractionary fiscal policy.

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LM04 Monetary Policy

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1. Introduction

This learning module covers:

- Roles and objectives of central banks
- Tools used to implement monetary policy and the monetary transmission mechanism
- Qualities of an effective central bank
- Monetary policy objectives: inflation, interest rate, exchange rate targeting
- Interaction of monetary and fiscal policy

2. Role of Central Banks

Roles of Central Banks and Objectives of Monetary Policy

- Monopoly supplier of the currency: Central banks are the only authority with the capacity to print money.
- Banker to the government and the banker's bank: Central banks provide banking services to the government and other banks in the economy.
- Lender of last resort: Because central banks have the capacity to print money, they are able to supply funds to banks that are facing a shortage and helps prevent a run on the bank.
- Regulator and supervisor of the payments system: Oversees, regulates and sets standards for a country's payments system for millions of transactions that happen on a daily basis. It also coordinates with other central banks around the world to formulate processes.
- Conductor of monetary policy: This is the most important role of the central bank. It takes actions to control or influence the quantity of money and credit in an economy.
- Supervisor of the banking system: This varies from one country to another. But, in many countries, the central bank along with one/more regulatory authorities oversees the banking system including granting licenses for a new bank, etc.
- Manage foreign currency reserves and gold reserves: The central bank may decide to sell foreign currency from its reserves if its domestic currency is under sustained pressure and continues to depreciate quickly. For example, The Reserve Bank of India (RBI) repeatedly sold dollars in 2012 to boost the Indian rupee.

The Objectives of Monetary Policy

Central banks normally have a variety of objectives (i.e., to maintain full employment and output, to maintain confidence in the financial system, or to promote understanding of the financial sector), but the overriding one is nearly always price stability or keeping inflation in check.

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3. Monetary Policy Tools and Monetary Transmission

The three tools available to central banks to control the money supply are: open market operations, refinancing rates and reserve requirements.

Open Market Operations

- Increase or decrease the amount of money in circulation.
- Increase money supply by buying government bonds from commercial banks → this increases the reserves of private banks → lends this money to firms and households, which then multiplies.
- Decrease money supply by selling government bonds to commercial banks → this decreases private banks' reserves to lend money to firms and households.

The Central Bank's Policy Rate

- Official interest rate: Known by different names in different countries. Called the official interest rate, or official policy rate, or policy rate, or discount rate. The objective is to influence short- and long-term interest rates.
- Repo rates: The rate at which the central bank is willing to lend money to commercial banks. The way it achieves this policy-rate setting is through a *repurchase agreement* wherein the central bank sells a security to commercial banks with a commitment to repurchase after a certain number of days. The maturity varies from overnight to two weeks; these are short-term collateralized loans. The rate implicit in this agreement is called repo rates.
- Base rates: The name varies from economy to economy. This is the rate at which commercial banks are willing to lend to each other.
- Federal funds rate**:** This is specific to the U.S. It is the interbank lending rate on overnight borrowings of reserves.
- In general, if the policy rate is high, the amount of lending will decrease and the quantity of money will decrease. On the other hand, if the policy rate is low, then the amount of lending will increase and the quantity of money will increase.

Reserve Requirements

The central bank may change the money supply in the economy by changing the reserve requirement. If the reserve requirement is low, then the money creation process is enhanced and the money supply increases. Conversely, if the reserve requirement is high, then the money creation process is restricted and the money supply decreases.

The Transmission Mechanism

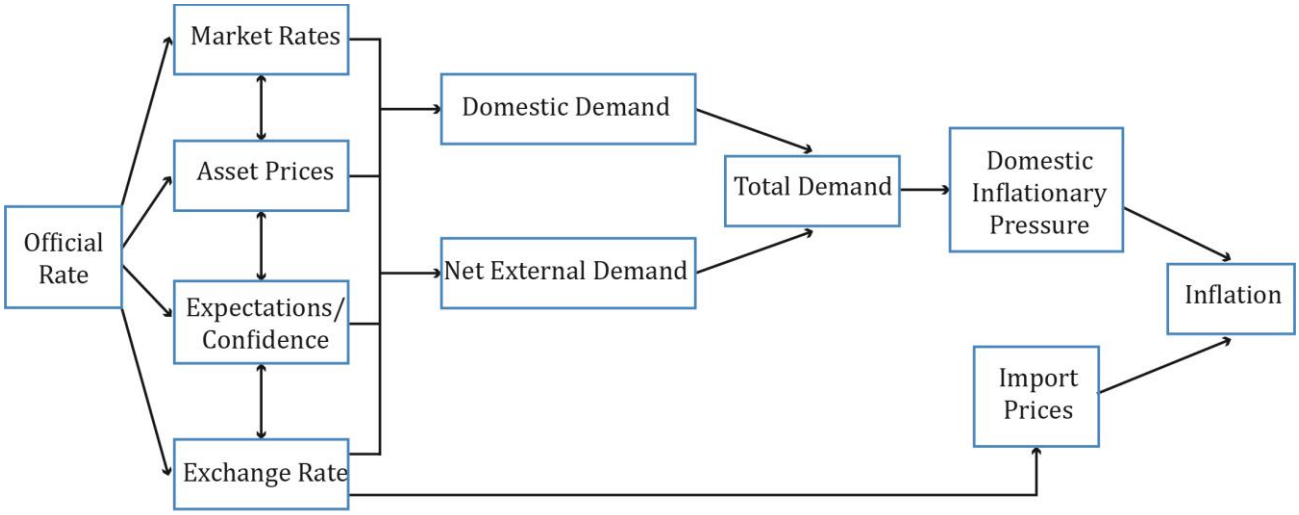
We looked at the tools used by central banks to influence rates, in the previous section. Now, we will see how policy decisions (especially the policy rate) transmits across the economy and affects the price level. One of the notable points from the previous section was that the

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policy rates set by central banks are short-term in nature, ranging from overnight to a few weeks. So, how does it affect the economy (growth, employment) in the long run?

The exhibit below is reproduced from the curriculum:

A Styled Representation of the Monetary Transmission Mechanism



Interpretation of the exhibit**:**

For simplicity, let us break it down into three parts.

1. Link between the changes in official rates on other related markets**:** To restrict money supply, the central bank may decide to increase official interest rates. This affects four interrelated channels:
   - *Market rates***:** An increase in official interest rates is reflected in short-term bank lending rates. Once the policy rate goes up, banks increase the base rates, which in turn affect the rates at which banks lend to customers (mortgages, loan rates). The rates on savings deposits also change, but not by the same amount as banks maintain a difference between deposit and loan rates.
   - *Asset prices*: Market value of securities such as bonds and equities decrease when the official interest rates rise. In the case of bonds, prices are inversely related to long-term interest rates. Securities prices also decrease because future cash flows are discounted by a larger factor.
   - *Expectations/confidence*: Increase in rates could influence (dent) expectations about the future prospects of the economy, such as employment opportunities, firms' profitability, unemployment levels, etc. A rate increase may also imply the economy is growing faster than expected. Or, it could mean it is time to slow the growth in the economy and tame inflation, which would in turn lower confidence.
   - *Exchange rate*: The exact impact is uncertain. But, an unexpected increase in official rate will lead to an appreciation of the domestic currency. A stronger domestic

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currency makes exports unattractive to overseas buyers as it is more expensive in their currency terms. This would lower the exporters' profitability.

Impact on individuals/households**:** Discourages borrowing, reduces spending, postponing consumption.

- Discourages consumer spending. Individuals tend to postpone consumption and are more inclined to save, if rates increase.
- The tendency to borrow to consume will also be low because of higher rates and interest rate expectations. If further rate increases are anticipated, then consumers will not borrow.
- Disposable incomes of individuals decrease as prices increase.
- The fall in asset prices affects their financial wealth. Higher rates → mortgage rates higher → lower demand → fall in market prices of houses.
- The outflow for those with mortgages increases.
- Makes domestic goods expensive relative to foreign goods. There may be a tilt in spending with more being spent on imported goods.

Impact on firms**:** Depends on the cost of capital

- Higher borrowing costs.
- Reduced profitability.
- Deferred new projects, investment spending, and hiring plans.

2. Impact on aggregate demand**:**

- Changes in consumption and investment spending behavior of consumers and firms affect aggregate demand. Low confidence → low spending → low demand.
- Increase in rate reduces real domestic demand and net external demand (difference between export and import consumption).
- Total aggregate demand decreases.

3. Impact on inflation**:**

- Determines the nominal value of goods and services in the long run.
- Weaker demand may put a downward pressure on inflation.

4. Monetary Policy Objectives

Inflation Targeting

Some economies implement monetary policy by targeting a certain level of inflation and then ensuring this level is met by monitoring a range of monetary and real economic indicators.

The success of inflation targeting depends on the following three factors that assess the effectiveness of a central bank:

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Central Bank Independence (free from political interference)	The degree of independence varies across economies. Some are operationally and target-independent, i.e., they determine the level of inflation to target, how to meet that target, and by when it must be achieved. There are others where the bank is assigned a rate of inflation to target by the government. Here, the government determines the appropriate level of inflation. So, they are only operationally independent.
Credibility	Follows through on its stated policy intentions. Is the central bank independent and does the market/public have confidence in the policy measures?
Transparency	Clear policy on economic indicators: Is the central bank transparent about its decision-making process? Being transparent in its quarterly assessment is one of the ways to gain credibility. What are the indicators the bank monitors before making the interest rate/policy decision?

Other features of an inflation-targeting framework include:

- A decision-making framework that considers a wide range of economic and financial market indicators.
- A clear, symmetric, and forward-looking medium-term inflation target, sufficiently above 0 percent, to avoid the risk of deflation, but low enough to ensure a significant degree of price stability. Similarly, it should not be too high either, because a high rate would result in price instability and inflation volatility.
- Exhibit 5 in the curriculum lists inflation targets for several economies. They are usually between 2% and 3%. For instance, if a country sets the target to 0.5% and misses it, then it runs the risk of deflation (or negative inflation).
- The inflation target set by central banks can become self-fulfilling prophecies if economic agents believe the target will be met. Wage negotiations will factor in this level of inflation.

The following are some of the obstacles in successful implementation of monetary policy in developing economies:

- Rapid financial innovation.
- Absence of liquid government-bond market.
- Lack of independence of central banks.
- Rapidly changing economy.
- Poor track record in controlling inflation in the past

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Exchange Rate Targeting

Instead of targeting inflation, some economies implement monetary policy by targeting the exchange rate. It is done by setting a fixed level or band of values for the exchange rate against a major currency.

How it works:

- Government/central bank announces the target.
- Central bank supports the target by buying and selling the national currency in foreign exchange markets.
- By tying the domestic economy's currency to another economy with a good track record of inflation, the domestic economy would effectively "import" the inflation experience of the low inflation economy.
- Interest rates and conditions in the domestic economy must adapt to accommodate this target and domestic interest rates and money supply can become more volatile.
    - Let us take a simple example: The central bank of Brazil announces that it wishes to maintain a specific exchange rate against the U.S. dollar. Brazil, being a developing economy, faces volatile inflation. If inflation is higher than the U.S.'s, then its currency, the Real, falls against the USD. To arrest its fall, Brazil's central bank would sell dollar reserves and buy Reals. It restricts the money supply and increases short-term interest rates. In contrast, if the inflation was low, and the Real appreciates against the USD, then the central bank would have to buy USD and sell Real. This explains how domestic interest rates and money supply can be volatile when targeting an exchange rate.

In a pegged exchange rate, a country fixes the value of its currency against either the value of another single currency, a basket of other currencies, or another measure of value, e.g., gold. In dollarization, a country uses US dollar as its functional currency.

What you cannot do because of exchange rate targeting:

- Monetary policy is not independent.
- Monetary policy cannot be used to target domestic inflation.

Contractionary and Expansionary Monetary Policies and Their Limitations

Central banks control liquidity by adjusting policy rates.

Contractionary monetary policy:

- High economic growth leads to high inflation. To cool off the economy, a central bank may employ a contractionary monetary policy.
- The central bank does this by increasing the official policy rate. This restricts the growth rate of money supply and the real economy contracts.

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Expansionary monetary policy:

- To boost a slowing economy, the central bank decreases the official policy rate. This increases liquidity and growth rate of money supply and the real economy expands.
- This is only in the short-run. Given the money neutrality theory, increasing or decreasing interest rates will not affect the real economy in the long-run.

High and low policy rate is with respect to the neutral rate of interest:

- It is the rate of interest that neither spurs nor slows the economy.
- Typically, neutral rate = Trend growth rate in the long run + long run expected inflation
- If policy rate > neutral rate, then the monetary policy is contractionary.
- If policy rate < neutral rate, then the monetary policy is expansionary.

Neutral rate = trend growth rate + inflation target

In an economy, if the trend growth rate is 3% and inflation target is 2%, then the neutral rate will be 5%. If the policy rate is set above 5%, then we have a contractionary policy. If the policy rate is set below 5%, then we have an expansionary monetary policy.

What's the Source of the Shock to the Inflation Rate?

The central bank must consider the source of inflation before deciding on contractionary/expansionary policy action. Two sources of shock to the inflation rate are:

- Demand shock: Caused by an increase in consumer confidence, which leads to more consumption and investment spending. Raising interest rates to control inflation is apt here.
- Supply shock: Caused by an increase in a supply factor such as oil prices. Raising interest rates is not appropriate as consumption will tend to fall, and consequently there will be an increase in unemployment.

Limitations of Monetary Policy

The will of the monetary authority does not necessarily transmit seamlessly through the economy.

This is because central banks cannot control:

1. The amount of money households and corporations put in banks on deposits.
2. The willingness of banks to create money by expanding credit.

It is relatively easy for central banks to influence short-term rates but long-term rates depend on expectations of interest rates and are not easy to control.

What is quantitative easing?

Quantitative easing is an unconventional monetary policy used when the traditional policy

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becomes ineffective. It is used to increase money supply where the central banks print (these days electronically) money to buy any assets.

5. Interaction of Monetary and Fiscal Policy

Both monetary and fiscal policies are used to stabilize an economy. But the impact of one varies based on the other's stance, and their interaction, as illustrated in the table below:

Note: convention in the table below

First line denotes the effect of fiscal policy

Second line denotes the effect of monetary policy

Third line denotes the overall effect on the economy

	Easy/Expansionary Fiscal Policy	**Tight Fiscal Policy**
Easy/Expansionary Monetary Policy	AD up. Low rates → private sector demand up. Growing private and public sector.	AD down. Low rates → private sector stimulated. The public sector will become a smaller percentage of the economy.
Tight Monetary Policy	AD up. High interest rates → private sector down. Public spending will become a higher percentage of GDP.	AD down. High interest rates → private sector demand down. Shrinking private and public sectors.

Factors Influencing the Mix of Fiscal and Monetary Policy

- To increase overall output, private investment spending is important. For this, monetary policy with low interest rates and tight fiscal policy works best.
- If infrastructure is poor and there is a lack of skilled labor, then an expansionary fiscal policy and loose monetary policy works best, but at the risk of inflation.
- Fiscal loosening methods that can be reversed after a specific time period include:
  - Social transfers to households.
  - Decrease in income tax.
  - Increase in government spending.
  - Increase in transfers to poor people.
- Monetary policies that will go with the above fiscal policies for the same duration include:
  - No monetary accommodation: any increase in AD will lead to increase in interest rates.

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- Interest rates stay unchanged for the duration when the fiscal policy is implemented and reversed.

Quantitative Easing and Policy Interaction

This section talks about quantitative easing – an unconventional approach adopted by the U.S. and U.K. governments during the recent recession of 2008-09 to print money. As interest rates were already near zero level, there was no option of reducing it further to stimulate growth. The central bank (printed money and) bought trillions of dollars' worth of government bonds to increase money supply in the system, increase expenditure, and avoid deflation.

The Importance of Credibility and Commitment

According to the IMF model, when governments run persistently large budget deficits, real interest rates rise, crowding out private investment and reducing each country's productive potential. As people realize that deficits will persist, inflation expectations and long-term interest rates rise, reducing the stimulus's effect by half.

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Summary

LO: Describe the roles and objectives of central banks.

Role of the central bank: supply currency; act as banker to the government and to other banks; regulate and supervise the payments system; act as a lender of the last resort; hold the nation's gold and foreign currency reserves; and conduct the monetary policy.

LO: Describe tools used to implement monetary policy tools and the monetary transmission mechanism, and explain the relationships between monetary policy and economic growth, inflation, interest, and exchange rates.

The three tools available to central banks to control the money supply are open market operations, policy rates, and reserve requirements.

Monetary transmission mechanism impacts in three ways: link between the change in official rates on other related markets, impact on aggregate demand, and impact on inflation.

The real trend rate is the long-term sustainable real growth rate of an economy.

The neutral interest rate is the sum of the real trend rate and the target inflation rate.

LO: Describe qualities of effective central banks; contrast their use of inflation, interest rate, and exchange rate targeting in expansionary or contractionary monetary policy; and describe the limitations of monetary policy.

Effective central banks exhibit the following: Independence (free from political interference), credibility (follow through on stated policy intentions) and transparency (clear policy on economic indicators).

Central banks can target a certain level of inflation and interest rate, and then ensure these levels are met by monitoring a range of monetary and real economic indicators.

Exchange rate targeting can be done by setting a fixed level or band of values for the exchange rate against a major currency.

Monetary policy is said to be contractionary when the policy rate is above the neutral rate and expansionary when the policy rate is below the neutral rate.

Monetary policy may not work as intended because central banks cannot control the amount of money households and corporations put in banks and the willingness of banks to create money by expanding credit.

Underdeveloped financial markets, rapid financial innovation, and lack of credibility of monetary authority may also hinder the utility of monetary policy.

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LO: Explain the interaction of monetary and fiscal policy.

	Easy/Expansionary Fiscal Policy	Tight Fiscal Policy
Easy/Expansionary Monetary Policy	AD up. Low rates → private sector demand up. Growing private and public sector.	AD down. Low rates→ private sector stimulated. The public sector will become a smaller percentage of the economy.
Tight Monetary Policy	AD up. High interest rates → private sector down. Public spending will become a higher percentage of GDP.	AD down. High interest rates → private sector demand down. Shrinking private and public sectors.

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1. Introduction

Geopolitics is the study of how geography affects politics and international relations. Under geopolitics, analysts study actors—the individuals, organizations, companies, and national governments that engage in political, economic, and financial activities—and how they interact with one another.

Geopolitical risk is the risk associated with tensions or actions between actors that affect the normal and peaceful course of international relations. Geopolitical risk could arise due to a change in policy, a natural disaster, a terrorist act, a theft, or war.

Investors study geopolitical risk because it can have a tangible impact on investment outcomes. This learning module provides a framework by which investors can measure, assess, track, and react to geopolitical risk, in order to improve investment outcomes.

2. National Governments and Political Cooperation

Actors can be split into two types:

- **State actors**: National governments, political organizations, or country leaders that exert authority over a country's national security and resources. For example, South African President, Malaysia's Parliament, and the British Prime Minister.
- **Non-state actors**: Entities that participate in global political, economic, or financial affairs but do not directly control national security or country resources. For example, NGOs, multinational companies, charities, and even influential individuals such as business leaders and cultural icons.

Features of Political Cooperation

At the highest level, the relationship between state actors can be cooperative or competitive.

Cooperation is the process by which countries work together toward some shared goal or purpose. These goals may vary widely from strategic or military concerns to economic influence or cultural preferences.

One specific type of cooperation – political cooperation – refers to the extent to which countries work toward agreements on rules and standardization for the activities and interactions with one another. We can classify countries as cooperative countries and non-cooperative countries.

- **Cooperative country**: A country that engages and reciprocates in rule standardization; tariff harmonization; international agreements on trade, immigration, or regulation; and allows free flow of information, including technology transfer.
- **Non-cooperative country**: A country with inconsistent and even arbitrary rules; restricted movement of goods, services, people, and capital across borders; retaliation; and limited technology exchange.

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Motivations for Cooperation

There are many reasons why a country may want to cooperate with its neighbors or with other state actors. These reasons are typically defined by a country's national interest.

National Security or Military Interest

National security (also known as national defense) is the protection of a country, its citizens, economy, and institutions from external threats. These threats can range from military attacks to terrorism, crime, cyber-security, and even natural disasters.

Geographic factors influence a country's approach to national security and the extent to which it will choose a cooperative approach.

- Landlocked countries, such as Switzerland, rely heavily on their neighbors for access to vital resources. Because of this reliance, cooperation is important for sustaining international access and growth, or even for survival.
- Countries that are well-connected to trade routes, such as Singapore, or countries acting as a conduit for trade, such as Panama, may use their geographic location as a lever of power.

Economic Interest

The concept of national security has evolved over time to include economic factors such as access to resources like energy, food, and water. Countries that choose to cooperate in support of their economic interests are most likely interested in one of two things: either securing essential resources through trade or levelling the global playing field for their companies or industries through standardization.

Resource Endowment, Standardization, and Soft Power

Geophysical Resource Endowment

Geophysical resource endowment includes factors such as livable geography and climate, as well as access to food and water, all of which are required for sustainable growth.

Geophysical resource endowment varies greatly among countries. Some countries (e.g. Unites States) are relatively self-sufficient in their resource use. While some countries (e.g. Japan) rely heavily on others for key factors of production, such as fossil fuels. These differences create power dynamics that can influence the terms of engagement between states.

Standardization

Standardization is the process of creating protocols for the production, sale, transport, or use of a product or service. Standardization can aid in the expansion of economic and financial activities across borders, benefiting all parties who agree to follow these protocols together.

The different types of rules standardization are: regulatory cooperation, process

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standardization, and operational synchronization. Exhibit 2 from the curriculum provides examples for each type.

	Regulatory Cooperation	**Process Standardization**	**Operational Synchronization**
Challenge	As financial cooperation expands, countries need a comprehensive standard for governance and risk management of the banking sector.	Financial transactions across borders faced higher costs and longer wait times, increasing the burden for cross-border activity.	Increasing international trade created supply chain bottlenecks as containers of different sizes and shapes were sent to ports worldwide.
Solution	Basel Committee on Banking Supervision (BCBS)	Society for Worldwide Interbank Financial Telecommunication (SWIFT)	Containerization
Process	The BCBS was established in 1974 by the G10 banking authorities. Membership has since expanded to the G20	SWIFT was established in 1973 to provide a global financial infrastructure.	Standards set for containers of uniform size and shape using multi-modal forms of transport (land, sea, air, rail) and port cranes.
Benefit	Allows for more effective supervision of the global banking sector and international capital flows.	Facilitates global payments in more than 200 countries and territories, servicing more than 11,000 institutions worldwide.	Dramatically reduces the time and cost of shipping goods.

Cultural Considerations and "Soft Power"

Some countries may cooperate with others for cultural reasons such as: long-standing political ties, immigration patterns, shared experiences, or cultural similarities.

Some countries may engage in soft power – which is a means of influencing another country's decisions without force or coercion. Soft power can be built over time through actions such as cultural programs, advertisement, travel grants, and university exchange. For example, South Korea promotes visiting Seoul (its capital) in subway systems around the world. These advertisements use popular Korean-made products, musical acts, and actors to encourage interaction with Korean culture and business.

The Role of Institutions

An institution is an established organization or practice in a society or culture. An institution can be a formal structure, such as a university, organization, or process backed by law; or, it

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can be informal, such as customs or behavioral patterns important to a society.

Strong institutions generally contribute to more stable internal and external political forces. This stability provides a country more opportunity to develop cooperative relationships. Stronger institutions also make cooperative relationships more durable.

Hierarchy of Interests and Costs of Cooperation

A country's national interest can be thought of as a hierarchy of factors. Those essential for survival are at the top and nice-but-not-essential elements are lower down.

Governments use the hierarchy of interests to guide their behavior. They will choose to cooperate where it benefits the nation-state, but when two needs result in conflicting cooperation tactics, those higher up the hierarchy take precedence. For example, two countries engaged in a military conflict may not cooperate on tariff harmonization, even though tariff harmonization may benefit both countries on a stand-alone basis.

Power of the Decision Maker

The length of a country's political cycle can influence its hierarchy of national interests. Many countries have political cycles of only a few years, making it difficult to prioritize long-term risks such as climate change or addressing income inequality over projects or goals that can be accomplished in a short period of time.

Decision makers, such as political parties or individuals, may have their own set of influences or needs, which can affect a country's cooperative and non-cooperative choices.

Political Non-Cooperation

Political cooperation and non-cooperation exist on a spectrum. While it is in some countries' interest to be highly politically cooperative, for others it is less essential. There are some cases of extreme non-cooperation—countries whose political autonomy is more important than the benefits of any cooperative actions.

3. Forces of Globalization

Globalization is the process of interaction and integration among people, companies, and governments worldwide. It has led to the spread of products, information, jobs, and culture across borders.

Features of Globalization

Globalization is primarily carried out by non-state actors, such as corporations, individuals, or organizations, and is the result of economic and financial cooperation.

Globalization is marked by economic and financial cooperation, including the active trade of goods and services, capital flows, currency exchange, and cultural and information exchange. By contrast, anti-globalization or nationalism is the promotion of a country's own economic interests to the exclusion or detriment of the interests of other nations. Nationalism is

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marked by limited economic and financial cooperation.

Motivations for Globalization

Globalization provides the following potential gains to non-state actors (such as companies and investors):

- **Increased profits**: Profits can be increased by increasing sales. Globalization allows companies to access new customers for their goods and services thereby increasing sales.

  Another way to boost profits is to reduce costs. Globalization allows companies to access lower-tax operating environments, reduce labor costs, and seek other supply chain efficiency gains, all of which are cost-reduction measures.

- **Access to resources and markets**: Companies need sustainable access to resources such as talent or raw materials. If these resources are not readily available or affordable in their home country, then companies may globalize to improve access.

  Some investors may seek investment opportunities in foreign markets. In this context, there are two important types of flows. Portfolio investment flows are short-term investments in foreign assets, such as stocks or bonds. Alternatively, foreign direct investments (FDI) are long-term investments in the productive capacity of a foreign country.

- **Intrinsic gain**: Intrinsic gain is a side effect or consequence of an activity that generates a benefit beyond profit itself. One example of intrinsic gain is the personal growth or education that people can obtain by broadening their horizons, visiting new places, or learning new ideas.

Costs of Globalization and Threats of Rollback

Globalization also has some potential disadvantages, such as:

- **Unequal accrual of economic and financial gains**: Although globalization generally improves aggregate economic activity, this does not imply improvement for everyone. For example, if a company relocates a factory to another country, it creates jobs in the new country but reduces jobs in the home country.

- **Lower ESG standards**: Companies that operate in lower-cost countries often adhere to the local standards of those countries. If the ESG standards in one country are lower as compared to another, and companies follow the lower standards, then globalization can create a drain on human, administrative, and environmental resources.

- **Political consequences**: The two costs listed above can lead to a third cost of globalization: the political consequences of global expansion. While some countries may benefit from increased labor force utilization, others may lose jobs as companies

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relocate. Therefore, globalization has the potential to exacerbate income and wealth inequality, as well as differences in opportunity within and between countries.

- **Interdependence**: Due to globalization companies may become dependent on other countries' resources for their supply chains. If there is a disruption to the supply chain, as witnessed in the COVID-19 pandemic, then companies may not be able to produce the good themselves.

4. International Trade Organizations

International Monetary Fund

Primary responsibilities of the IMF: To ensure the stability of the international monetary system, the system of exchange rates, and international payments that enables countries to buy goods and services from each other.

- Provides a forum for cooperation on international monetary problems.
- Facilitates the growth of international trade and promotes employment, economic growth, and poverty reduction.
- Supports exchange rate stability and an open system of international payments.
- Lends foreign exchange to members when needed, on a temporary basis and under adequate safeguards, to help them address balance of payments problems.

How does the IMF ensure global economic stability?

- **Financial assistance**: The IMF stands ready to lend foreign currencies to member countries to assist them during periods of significant external deficits. A pool of gold and currencies contributed by members provides the IMF with the resources required for these lending operations. Following the 2007-09 crisis, the IMF has enhanced its lending facilities and member countries' access to fund resources.
- **Surveillance**: In the wake of several financial crises in the recent past (downgrading of Greek sovereign debt to non-investment grade and its cascading effects on other EMU countries such as Italy, Spain, etc.), the IMF has improved its monitoring of global, regional, and country economies on macroeconomic policies.
- Helping resolve global economic imbalances.
- **Technical assistance:** Offers training to country officials on how to design and implement effective macroeconomic policies, and how to manage their financial systems and capital markets.

World Bank Group

The World Bank's main objective is to help developing countries fight poverty and enhance environmentally sound economic growth. One of the current goals of the World Bank to achieve by 2030 is to end extreme poverty by decreasing the percentage of people living on less than $1.25 a day to less than 3%. (Source: worldbank.org)

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The World Bank funds projects in developing countries and provides financial and technical expertise with an objective to reduce poverty. It finances these projects through its two affiliated entities, IBRD and IDA. Both these organizations provide low or no interest loans to countries that have unfavorable or no access to international credit markets.

- **The International Bank for Reconstruction and Development** (IBRD): The primary source of funding for the bank is by selling AAA- rated bonds in the world's financial markets. It has enjoyed this high rating since 1959. IBRD's reserves have been built up over the years with the capital contributed by the bank's 188 member countries. It generates income from lending out its own capital. IBRD's income pays for the World Bank's operating expenses.

The International Development Association (IDA): The world's largest source of interest-free loans and grant assistance to the poorest countries.

World Trade Organization

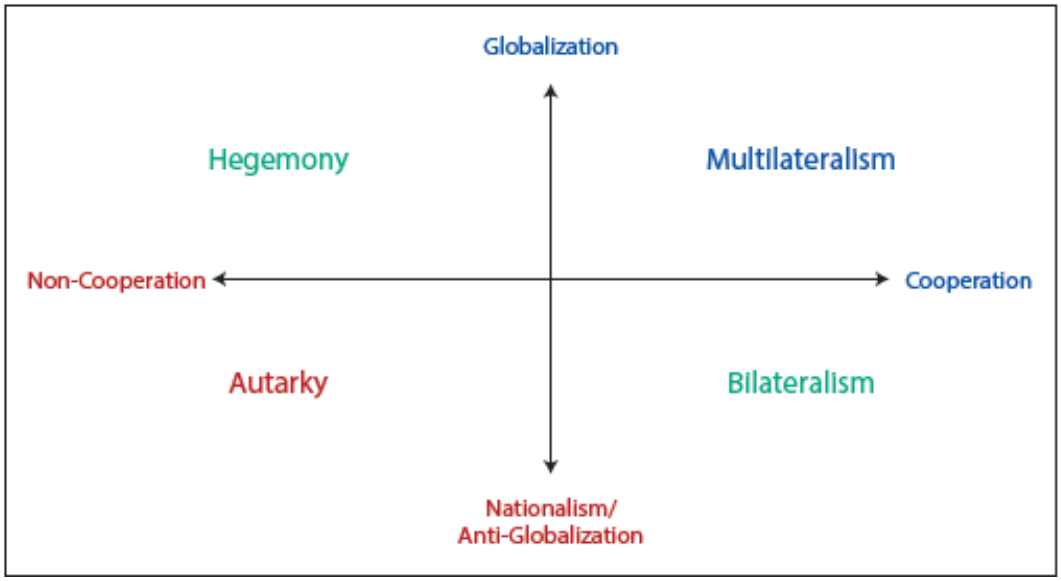
The WTO is the only organization that deals with the rules of trade between nations. The most important functions of the WTO include:

- Implementation, administration, and operation of individual agreements. All the major decisions are taken by the representatives of the governments who meet regularly in Geneva.
- Acting as a platform for negotiations such as lowering customs tariffs, removing trade barriers, etc.
- Settling disputes: If the countries feel there has been an infringement of an agreement, or any other dispute, then the issue is settled by the WTO.
- Building trade capacity: The WTO helps developing countries to build the skills and infrastructure needed to boost their trade.
- WTO agreements have been signed by a large majority of the world's trading nations and ratified in their respective parliaments.
- The WTO has the mandate to review and propagate its members' trade policies and ensure the coherence and transparency of trade policies through surveillance in a global policy setting.

5. Assessing Geopolitical Actors and Risk

Geopolitical actors can be classified into four categories based on two axes – 'political cooperation versus non-cooperation' and 'globalization versus nationalism'. Exhibit 7 from the curriculum illustrates this classification.

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The four categories are: autarky, hegemony, multilateralism, and bilateralism. Each category has its own costs, benefits, and tradeoffs with respect to geopolitical risk.

In general terms, regions, countries, and industries that are more reliant on on cross-border goods and capital flows will have higher levels of cooperation.

The interdependent nature of globalization may reduce the likelihood that collaborative countries levy economic, financial, or political attacks on one another. However, that same interdependence can make cooperative actors more vulnerable to geopolitical risk than those less dependent on cooperation and trade.

For geopolitical risk analysis, it is important to consider not only which quadrant a country currently occupies, but also its stability within that quadrant. A hegemon working to strengthen political cooperation may pose less of a threat to investment outcomes than a multilateral actor attempting to undermine them.

Autarky

Autarky refers to countries that seek political self-sufficiency with little or no external trade or finance. Strategic domestic industries are controlled by state-owned enterprises. The self-sufficiency of autarkic countries allows them to be stronger politically. They are able to exercise complete control over the supply of technology, goods and services, as well as media and political messaging.

In some cases, autarky can allow a country to achieve rapid economic and political development. For example, during much of the 20th century China acted as an autarkic. During this period, China was able to reduce poverty significantly and has since moved towards more economic and financial cooperation.

However, in other cases, such as North Korea and Venezuela, autarky has resulted in a gradual loss of economic and political development within the country.

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Hegemony

Hegemonic countries tend to be regional or even global leaders, who use their political or economic influence over others to control resources.

A hegemonic system can benefit both the hegemonic countries themselves and the international system. The economic and political dominance of a hegemon may provide it the ability to significantly influence global affairs. Countries that follow the hegemon's rules and standards may reap the benefits provided by the leader, including the stabilizing force of monitoring and enforcing the hegemon's standards.

However, hegemonic systems also have costs. Hegemons may become more competitive as they gain or lose influence, increasing the likelihood of geopolitical risk.

Example of hegemonic countries include the United States and Russia.

Multilateralism

Multilateralism refers to countries that participate in mutually beneficial trade relationships and extensive rule harmonization. Private firms are fully integrated into global supply chains with multiple trade partners.

Examples of multilateral countries include Germany and Singapore. Factors that have led Singapore to rely heavily on globalization to achieve economic success are:

- Factor endowment: Singapore has limited natural resources, including water and arable land. This makes it highly dependent on trade partners and innovation to survive.
- Geographic location: Singapore is located at the intersection of many important global trade routes which helps make it an Asian center for world business.
- Cultural factors: Singapore's population is highly ethnically and racially diverse, fluent in English and highly skilled.
- Institutional factors: Singapore has highly stable political institutions with high governmental priorities on promoting economic activity and enforcing business-friendly institutional governance. This makes it a preferred investment destination and an attractive trade partner.

Bilateralism

Bilateralism is the conduct of political, economic, financial, or cultural cooperation between two countries. Countries that engage in bilateralism may have relations with many different countries, but these are one-at-a-time agreements without multiple partners.

Countries typically exist on a spectrum between bilateralism and multilateralism. Between the two extremes is **regionalism**, which involves a group of countries collaborating with one another. For example, regional blocs may agree to provide trade benefits to one another while raising barriers for those outside the group.

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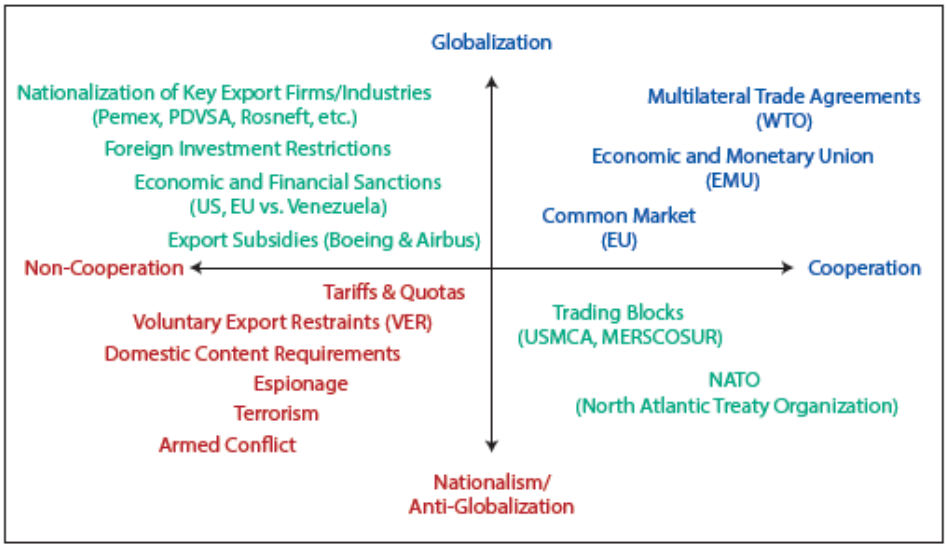
Japan once fit the description of bilateralism perfectly. Its government engaged in significant political cooperation in order to develop a strong export market, but it did not globalize its imports. However, today Japan would be considered a multilateral player.

6. The Tools of Geopolitics

Geopolitical tools can be classified into three types:

- National security tools
- Economic tools
- Financial tools

As discussed in Sec 5, geopolitical risk analysts must monitor not only which quadrant best describes an actor, but also its stability within that quadrant. The use of various national security, economic, or financial tools may indicate that an actor's character is changing, which may increase or decrease geopolitical risk. Exhibit 9 from the curriculum depicts how geopolitical tools fit into the four archetypes of country behavior.



National Security Tools

National security tools are those used to influence or coerce a state actor through direct or indirect impact on the country's resources, people, or borders. National security tools may be active (in use at the time of analysis), or threatened (likely to be used in near future).

Armed conflict is the most extreme example of a national security tool. It is a direct and active national security tool that can have two major consequences:

- Destruction of physical infrastructure, which can have a long-term impact on a country's capital stock and ability to rebuild that stock.
- Migration away from areas of armed conflict, which has the potential to alter international flows of goods, services, capital, and labor. It may also impact neighboring countries and states that accept refugees.

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Not all national security tools used are as direct in nature as armed conflict. For example, *espionage* (the practice of using spies to obtain political or military information) is an indirect national security tool. Another example is *miliary alliances* which are frequently used to either aid in direct conflict or to prevent conflict from occurring in the first place.

Also, not all national security tools are necessarily used in a non-cooperative way. For example, the North Atlantic Treaty Organization (NATO), an alliance between the European Union, United States, United Kingdom, and Canada which was originally constructed to provide collective security against the Soviet Union, is now used to discuss and de-escalate potential conflict among members and between members and outside states.

Economic Tools

Economic tools are the actions used to reinforce cooperative or non-cooperative stances via economic means.

Examples of cooperative economic tools include *multilateral trade agreements* (such as Southern Common Market - MERCOSUR, which facilitates trade among member countries in South America) and the *global harmonization of tariff rules* (as facilitated by the World Trade Organization, WTO).

Examples of non-cooperative economic tools include *nationalism* (the process of transferring an activity or industry from private to state control), *voluntary export restraints* (refusing to trade as much of their goods and services as would meet demand), and imposing *domestic content requirement* (stating that a certain proportion of domestic input be included in an exported good.)

Financial Tools

Financial tools are the actions used to reinforce cooperative or non-cooperative stances via financial mechanisms.

Examples of cooperative financial tools include the *free exchange of currencies* across borders and *allowing foreign investment*.

Examples of non-cooperative financial tools include *limiting access to local currency markets* and *restricting foreign investment*.

Cooperative financial tools that encourage cooperation in security, economic, and financial arenas may reduce geopolitical risk. However, if the system becomes too dependent on a particular financial tool, it may introduce vulnerabilities in the international system that can have far-reaching consequences. For example, transactions in the international interbank market are heavily denominated in US dollars. While the market itself represents a form of cooperation, over reliance on the US dollar makes other countries vulnerable to changes in the US monetary policy.

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Multifaceted Approaches

The national security, economic, and financial geopolitical tools discussed above can be intertwined. An example is *cabotage* which refers to the right to transport passengers or goods within a country by a foreign firm. Many countries, including those with multilateral trade agreements, impose cabotage restrictions across transportation subsectors. This means that shippers, airlines, and truck drivers are not permitted to transport goods and services within the borders of another country. Allowing cabotage requires multilateral coordination in areas such as physical security and economic coordination.

Geopolitical Risk and Comparative Advantage

Geopolitical risk and tools of geopolitics can affect the comparative advantage of a country. Countries with low geopolitical risk may attract more labor and capital. Whereas, countries with high geopolitical risk may suffer a loss of labor and capital. For instance, Germany's reaction to the Syrian refugee crisis is an example of comparative advantage stemming from geopolitical risk. Germany was able to resettle one million Syrian refugees due to its strong economic position in the EU and long-standing political stability. By adding young and talented migrants, Germany's long-term demographic balance has improved which can potentially increase its economic growth rate.

7. Geopolitical Risk and the Investment Process

The degree to which investors consider geopolitical risk in their decision-making will vary widely depending on their investment objectives and risk tolerance. Some investors may not consider geopolitical risk separately, whereas others may make geopolitical risk a central component of their investment strategy.

Types of Geopolitical Risk

The three basic types of geopolitical risk are:

- **Event risk:** Event risk evolves around set dates, such as elections, new legislation, or other date-driven milestones, such as holidays or political anniversaries, known in advance. Political events often cause investors to change their expectations about a country's cooperative stance. Brexit is an example of event risk.
- **Exogenous risk:** Exogenous risk is a sudden or unanticipated risk that impacts either a country's cooperative stance, the ability of non-state actors to globalize, or both. Examples include sudden uprisings, invasions, or the aftermath of natural disasters.
- **Thematic risk:** Thematic risks are known risks that evolve and expand over a period of time. Examples include climate change, cyber threats, and the ongoing threat of terrorism.

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Assessing Geopolitical Threats

To assess geopolitical threats, an investor considers geopolitical risk in terms of the following three areas:

- **Likelihood it will occur**: Refers to the probability that a risk will occur. Measuring the likelihood is a challenging process and this exercise can be more art than science.
- **Velocity (speed) of its impact**: Refers to the pace at which a risk may impact an investor's portfolio. Impacts can be classified into:
    - Short-term or high velocity impacts: Exogeneous or 'black swan' (an event that is rare and difficult to predict but has a significant impact) events fit this category. Such events cause market volatility and investor flight to quality in the short term. But long-term changes may not be required.
    - Medium-term impacts: Risks with medium term impact may affect a company's process, costs, and investment opportunities, resulting in lower valuations. These risks tend to impact some companies/sectors more than others.
    - Long-term or low velocity impacts: Long-term risks may have significant ESG impacts requiring an adjustment to an investor's long term asset allocation. However, they may have limited immediate impact on portfolios.
- **Size and nature of that impact**: A risk's impact on a portfolio can manifest in many different ways. Generally, investors are more concerned with risks that have a high-impact and may ignore risks that have a low-impact.

  Also, the size of a risk's impact can be magnified by external factors. For example, risk tends to have a greater impact on markets experiencing an economic downturn.

  Impacts may be discrete (affecting only one company/sector) or broad (affecting the entire economy) in nature. For example, cyber risks can have both discrete and broad impacts. In the event of a cyberattack, the specific company will be impacted. It will also impact all other companies by increasing their monitoring, due diligence, and security costs.

Geopolitical risks rarely evolve in a linear fashion, which makes it difficult to forecast their likelihood, velocity, size and nature of impact on a portfolio. Therefore, instead of using a single point forecast, many investors use a scenario building and signposting approach.

Scenario analysis is the process of comparing portfolio outcomes across different scenarios or states of the world. Scenarios can be qualitative, quantitative, or a combination of the two.

Tracking Risks According to Signposts

A signpost is an indicator, market level, data piece, or event that signals a risk is becoming more or less likely. We can think of signposts like a traffic light.

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- If the risk is low in likelihood, velocity, or impact then the signposts are flashing 'green' indicating 'no action needed'.
- If the risk is medium in likelihood, velocity, or impact then the signposts are flashing 'amber' indicating 'higher caution and preparedness'.
- If the risk further rises in likelihood, velocity or impact then the signposts are flashing 'red' indicating 'an action plan is necessary'.

Consider the 'Brexit' example. Before 2014, the signposts for geopolitical risk in the UK may have flashed green. When the referendum was announced in 2015, the signposts flashed amber. In May 2016, when phone polls suggested that the "leave" vote was gaining majority, the signposts flashed red.

Manifestations of Geopolitical Risk

High-velocity risks are most likely to manifest themselves in market volatility through sudden changes in asset prices. For example, in response to the COVID-19 pandemic, the S&P 500 index declined nearly 34%. Also, investors flight to safety pushed up US bond prices and the US 10-year treasury yield declined by nearly 68%. Note, however that this impact was not permanent and the markets eventually recovered.

Low-velocity geopolitical risks can have a more prolonged impact. Sustained disruption can lead to lower revenues, higher costs, or both, all of which can have a negative impact on a company's valuation.

Investors may require higher compensation for countries, regions, or sectors perceived to be at higher risk of geopolitical disruption, effectively increasing the discount rate investors use when valuing those securities. This is a key reason why investors demand a higher required return for investments in emerging and frontier markets as compared to developed markets.

Acting on Geopolitical Risk

On a macroeconomic level, geopolitical risks can impact capital market conditions, such as economic growth rates, interest rates, and market volatility. This in turn can influence the asset allocation decisions of investors.

Geopolitical risks can also influence the amount of capital an investor allocates to different countries. For example, more capital will be allocated to countries with a long history of using a multilateral approach and less capital will be allocated to countries experiencing consistent military threat.

On a portfolio level, investors can consider geopolitical risk as one of the factors in a multi-factor model. Geopolitical risk can influence the appropriateness of an investment security or strategy for an investor's goals, risk tolerance, and time horizon. For example, an investor with low risk tolerance should reduce exposure to geopolitical risk through low-volatility investment choices or through hedging.

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Summary

LO. Describe geopolitics from a cooperation versus competition perspective.

Geopolitics is the study of how geography affects politics and international relations. Under geopolitics, analysts study actors—the individuals, organizations, companies, and national governments that engage in political, economic, and financial activities—and how they interact with one another.

Political cooperation refers to the extent to which countries work toward agreements on rules and standardization for the activities and interactions with one another. We can classify countries as cooperative countries and non-cooperative countries.

- Cooperative country: A country that engages and reciprocates in rule standardization; tariff harmonization; international agreements on trade, immigration, or regulation; and allows free flow of information, including technology transfer.
- Non-cooperative country: A country with inconsistent and even arbitrary rules; restricted movement of goods, services, people, and capital across borders; retaliation; and limited technology exchange.

Political cooperation and non-cooperation exist on a spectrum. While it is in some countries' interest to be highly politically cooperative, for others it is less essential.

LO. Describe geopolitics and its relationship with globalization.

Globalization is primarily carried out by non-state actors, such as corporations, individuals, or organizations.

Globalization is marked by economic and financial cooperation, including the active trade of goods and services, capital flows, currency exchange, and cultural and information exchange. By contrast, anti-globalization or nationalism is the promotion of a country's own economic interests to the exclusion or detriment of the interests of other nations. Nationalism is marked by limited economic and financial cooperation.

Geopolitical actors can be classified into four categories based on two axes – 'political cooperation versus non-cooperation' and 'globalization versus nationalism'. The four categories are: autarky, hegemony, multilateralism, and bilateralism. Each category has its own costs, benefits, and tradeoffs with respect to geopolitical risk.

LO: Describe functions and objectives of the international organizations that facilitate trade, including the World Bank, the International Monetary Fund, and the World Trade Organization.

International Monetary Fund:

- International monetary cooperation and exchange rate stability.
- Assists in setting up international payments systems; makes resources available to member countries with balance of payments problems.

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World Bank:

- Low-interest loans, interest-free credits, and grants to developing countries for many specific purposes.
- Provides resources and knowledge and helps form private/public partnerships with the overall aim of fighting poverty.

World Trade Organization:

- Ensures that trade flows freely and works smoothly.
- Main focus: Institute, interpret, and enforce a number of multilateral trade agreements which detail global trade policies for a large majority of the world's trading nations.

LO: Describe geopolitical risk.

The three basic types of geopolitical risk are:

- Event risk: Evolves around set dates known in advance.
- Exogenous risk: A sudden or unanticipated risk that impacts either a country's cooperative stance, the ability of non-state actors to globalize, or both.
- Thematic risk: Known risks that evolve and expand over a period of time.

LO. Describe tools of geopolitics and their impact on regions and economies.

Geopolitical tools can be classified into three types:

- National security tools: e.g., armed conflict, espionage, military alliances
- Economic tools: e.g., multilateral trade agreements, global harmonization of tariff rules
- Financial tools: e.g., exchange of currencies across borders, foreign investments

Geopolitical tools can be used in both a cooperative or non-cooperative way.

LO. Describe the impact of geopolitical risk on investments.

Investors study geopolitical risk because it has a tangible impact on investment outcomes.

To assess geopolitical threats, an investor considers geopolitical risk in terms of the following three areas:

- Likelihood it will occur
- Velocity (speed) of its impact
- Size and nature of that impact

Geopolitical risks rarely evolve in a linear fashion, which makes it difficult to forecast their likelihood, velocity, size and nature of impact on a portfolio. Therefore, instead of using a single point forecast, many investors use a scenario building and signposting approach.

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LM06 International Trade

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1. Introduction

This learning module covers:

- Benefits and costs of international trade
- Types of trade restrictions and their economic implications
- Trading blocs and regional integration

2. Benefits and Costs of Trade

Benefits of international trade:

- Countries gain from the exchange and specialization. For example, if one country is good at producing cloth, while another is good at manufacturing machinery, then the two countries can exchange the goods. A country benefits from exchange trade if either of the following two conditions occur in an exchange, (as there is efficient allocation of resources):
  - Higher price for exports relative to selling internally.
  - Lower price for imports relative to producing internally.
- Households and firms have greater product variety. For example, after South Korean electronics manufacturers such as LG and Samsung entered the South Asian markets, consumers benefited because they had access to a wide range of products. This argument can be extended to almost any industry from automobiles to food products.
- Competition increases and resources are allocated more efficiently.
  - In an open economy, increased competition from foreign forms reduces the monopoly power of domestic firms and forces them to be efficient.
  - By increasing production of the exported good and reducing production of the imported good, a country achieves a more efficient allocation of resources. This efficiency enables the consumption of a larger basket of goods, increasing overall welfare.
- Industries experience greater economies of scale.
  - In an open economy, companies are forced to compete with global players and that forces them to become more efficient or go out of business. Extending the LG and Samsung example from above, economies of scale allows these firms to benefit from the larger market size in India/Middle East and experience lower average cost of production.
- Greater employment in exporting countries.
  - The garment factories in Bangladesh export cotton clothing to the rest of the world for various labels. The work in this industry is labor-intensive and hence generates a lot of employment opportunities.

Note: The term "gains from trade" implies that the overall benefits of trade outweigh the losses from trade. It does not mean that all stakeholders (producers, consumers, government) benefit (or benefit equally) from trade.

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Costs of international trade:

- Potential income inequality.
  - It depends on whether the industry is expanding or contracting. For example, the IT industry in the late 90's and early 2000's created high-paying jobs. This led to an income disparity between the IT industry and other traditional industries such as automobiles.
- Loss of jobs because less inefficient firms will be forced to exit.
  - For example, the U.S. textile industry has faced tough competition because of cheaper imports from other countries forcing many companies to exit.

3. Trade Restrictions and Agreements—Tariffs, Quotas, and Export Subsidies

Trade restrictions are government policies that limit the ability of domestic households and firms to freely trade with other countries.

Tariffs

Tariffs are taxes imposed by a government on imported goods. Tariffs are common in countries where the government finds it difficult to collect taxes from its citizens.

Why governments levy tariffs:

- To protect domestic industries.
- To reduce trade deficit. Tariffs reduce the demand for imports by increasing their price above the free trade price.

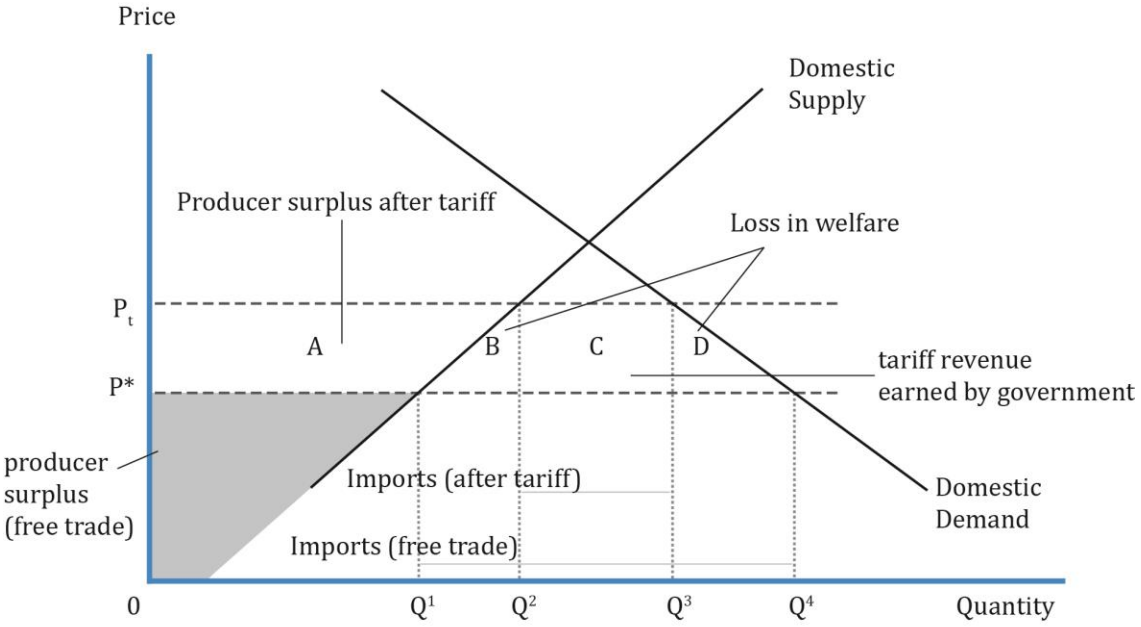
"Small" country: A country is a price taker in the world market for a product, and is too small to affect the world market price. Small country does not mean it is small in size, population, or GDP. For example, India is a large country, but it is a price taker for luxury bikes like Ducati, Harley Davidson, and Triumph.

"Large" country: A country is a large importer of the product and can influence the world market price. For example, the United States is a large country for the automobile market. If it were to impose tariffs on imported cars, then exporters would reduce the price of the cars to retain market share.

Generally, tariffs result in loss in global welfare. The exhibit below illustrates the wealth effects of tariffs and quota.

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Welfare effects of Tariff and Import Quota:



Key points of the exhibit are summarized below:

- Under free trade: Q^1 is domestic supply, Q^4 is domestic consumption. Import demand = demand – supply = the distance between Q^1 and Q^4. Assume the country is Portugal; a small country from a trade perspective, and it imports cars. It is a price taker and the price per car (P^*) is 100.
- After tariff is imposed: Q^2 is domestic supply, Q^3 is domestic consumption. Import demand = the distance between Q^2 and Q^3. Domestic producers supply more and consumers demand less. The price is represented byP^t. Assume Portugal imposes a 50% tariff, so the price of the car is now 150.

Interpretation of what happens after the tariff:

- Tariff results in a deadweight loss, known as welfare loss here, denoted by B + D.
- Producer surplus increases to A because of a higher price for their output.
- Government revenue increases because of the tariff collected, as denoted by C.
- Consumer surplus decreases because of the increase in price.

The welfare effects are tabulated below:

	Importing country
Consumer surplus	- (A + B + C + D)
Producer surplus	+ A
Tariff revenue or quota rents	+ C
National welfare	- B - D

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This example was for a small country. Now, let us look at it from a large country's perspective. The terms of trade change. Assume the large country is the United States that imports cars from Japan. If the U.S. imposes a tariff, then Japan will reduce the price of cars. Terms of trade for the U.S. will improve as the price of imports has gone down.

The outcome is:

- For the importing country (U.S.): producers gain, but consumers lose.
- Creates tariff revenue for the government.
- For the exporting country (Japan): producers lose but consumers gain.

Example: Analysis of a Tariff
(*This is Example 1 from the curriculum.*)

South Africa manufactures 110,000 tons of paper. However, domestic demand for paper is 200,000 tons. The world price for paper is USD5.00 per ton. South Africa will import 90,000 tons of paper from the world market at free trade prices. If the South African government (a small country) decides to impose a tariff of 20 percent on paper imports, the price of imported paper will increase to USD6.00. Domestic production after the imposition of the tariff increases to 130,000 tons, while the quantity demanded declines to 170,000 tons.

1. Calculate the loss in consumer surplus arising from the imposition of the tariff on imported paper.

Solution:

The loss in consumer surplus = USD1 × 170,000 + 1/2 × USD1 × 30,000 = USD185,000. This calculation is represented by areas A + B + C + D.

2. Calculate the gain in producer surplus arising from the imposition of the tariff.

Solution:

Gain in producer surplus = USD1 × 110,000 + 1/2 × (USD1 × 20,000) = USD120,000; Area A.

3. Calculate the gain in government revenue arising from the imposition of the tariff.

Solution:

Change in government revenue = USD1 × 40,000 = USD40,000; Area C.

4. Calculate the deadweight loss arising from the imposition of the tariff.

Solution:

Deadweight loss because of the tariff = 1/2 × USD1 × 20,000 + 1/2 × USD1 × 30,000 = USD 25,000; Areas B + D.

Quotas

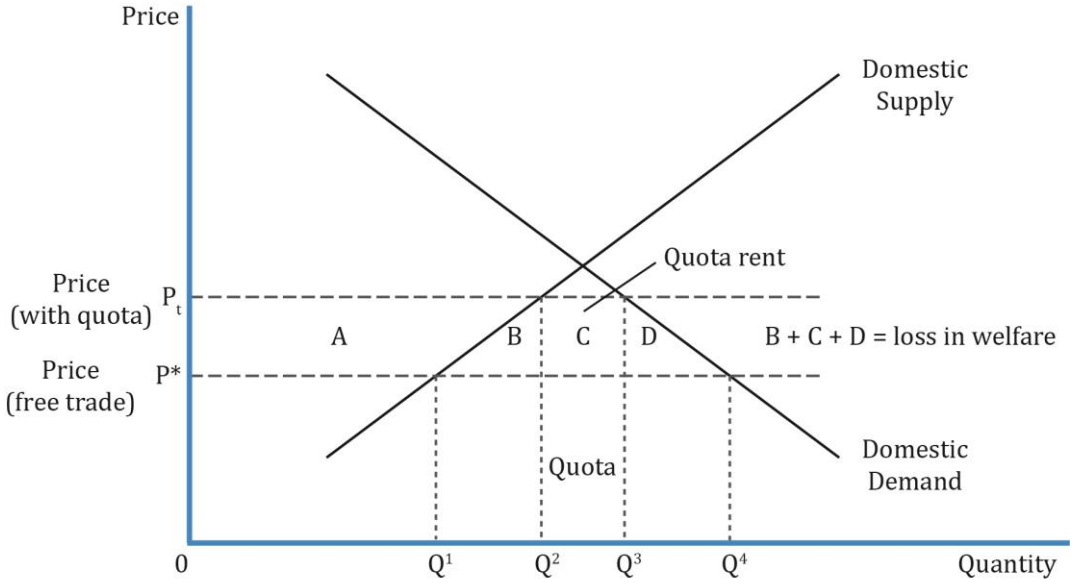
Quota is a restriction on the absolute amount (quantity) of imports allowed over some

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period, typically a year. An *import license* specifies the maximum quantity that can be imported during a given period; it is used to implement a quota.

- Quota rents: The extra profit earned by foreign producers. Exporters earn greater profits with quotas because they often raise the price of their goods.
- To understand the difference between tariffs and quotas, let us consider the same example.

Welfare effects of Tariff and Import Quota



Interpretation:

- At first glance, tariff and quota look similar. But there is a subtle difference between the two in government revenue and loss of welfare. With tariffs, it was clear that region C was the revenue earned by the government. But, in the case of quota, these are profits earned by the exporters as they raise the prices of their goods. If there was no quota, the prices of these goods would not have increased.
- With tariffs, the loss in welfare was equal to the deadweight loss of B+D. But, with quotas, it is equal to B+C+D. The amount lost in C can be eliminated if the government sells the import licenses to the exporters for a fee, and this amount must be equal to C.

Voluntary export restraint: Export quota administered by the exporting country; exporting country agrees to limit exports of a particular good usually at the request of the importing country to avoid tariffs or quotas. A VER allows importing countries to protect domestic industries from a surge of imports. The difference between an import quota and a VER is that the former is imposed by the importing country, while the latter is imposed by the exporter. One example of a VER is between Japan and U.S. from 1981 to 1994. The U.S. recognized the rising popularity of Japan's cars in the early 1980s and wanted to protect its domestic automobile industry. In 1981, the Japanese government responded by entering into a VER

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agreement with the U.S. limiting the number of cars exported to the United States to 1.68 million a year.

Export Subsidies

Export subsidies are payments by the government to a firm for each unit exported:

- The objective is to stimulate exports, increase production in certain industries, and create domestic employment.
- The exporter has an incentive to focus on the export market because the firm receives the international price plus the per-unit subsidy for each unit of the good exported.
- The most export-subsidized industry in the world is agriculture.
- Countervailing duties are duties levied by the importing country against subsidized exports entering the country. This tariff is imposed to offset the effect of subsidy.
- If a small country imposes export subsidies, domestic price rises.
- If a large country imposes export subsidies, world prices decline as quantity increases.
- For example, the European Union subsidizes sugar and is the second largest exporter in the world.
- Net welfare is down in the large and small country.

The exhibits below are reproduced from the curriculum.

Instructor's Note: these are important and testable. They summarize the effects of all the alternative trade policies we have seen so far.

Effects of Alternative Trade Policies				
	Tariff	**Import quota**	**Export subsidy**	**VER**
Impact on	Importing country	Importing country	Exporting country	Importing country
Producer surplus	Increases	Increases	Increases	Increases
Consumer surplus	Decreases	Decreases	Decreases	Decreases
Government revenue	Increases	Mixed (depends on whether the quota rents are captured by the importing country through the sale of licenses, or by the exporters)	Falls (government spending rises)	No change (rent to foreigners)
National welfare	Decreases in small country	Decreases in small country	Decreases	Decreases

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	(could increase in large country).	(could increase in large country).		

Effects of alternative trade policies on price, production, consumption, and trade				
	Tariff	**Import quota**	**Export subsidy**	**VER**
Impact on	Importing country	Importing country	Exporting country	Importing country
Price	Increases	Increases	Increases	Increases
Domestic consumption	Decreases	Decreases	Decreases	Decreases
Domestic production	Increases	Increases	Increases	Increases
Trade	Imports decrease	Imports decrease	Exports increase	Imports decrease

Example: Analysis of Trade Restrictions
(*This is Example 2 from the curriculum.*)

Thailand, a small country, has to decide whether to impose a tariff or a quota on the import of computers. You are considering investing in a local firm that is a major importer of computers.

1. What will be the impact of a tariff on prices, quantity produced, and quantity imported in Thailand (the importing country)?

Solution:

A tariff imposed by a small country, such as Thailand, raises the price of computers in the importing country, reduces the quantity imported, and increases domestic production.

2. If Thailand imposes a tariff, what will the impact be on prices in the exporting country?

Solution:

A tariff imposed by a small country would not change the price of computers in the exporting country.

3. How would a tariff affect consumer surplus, producer surplus, and government revenue in Thailand?

Solution:

When a small country imposes a tariff, it reduces consumer surplus, increases producer surplus, and increases government revenue in that country

4. Explain whether the net welfare effect of a tariff is the same as that of a quota.

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Solution:

The quota can lead to a greater welfare loss than a tariff if the quota rents are captured by the foreign government or foreign firms.

5. Which policy, a tariff or a quota, would be most beneficial to the local importer in which you may invest and why?

Solution:

A tariff will hurt importers because it will reduce their share of the computer market in Thailand. The impact of a quota depends on whether or not the importers can capture a share of the quota rents. Assuming importers can capture at least part of the rents, they will be better off with a quota.

6. If Thailand were to negotiate a VER with the countries from which it imports computers, would this be better or worse than an import quota for the local importing firm in which you may invest? Why?

Solution:

The VER would not be better for the local importer than the import quota and would most likely be worse. Under the VER, all of the quota rents will be captured by the exporting countries whereas with an import quota at least part of the quota rents may be captured by local importers.

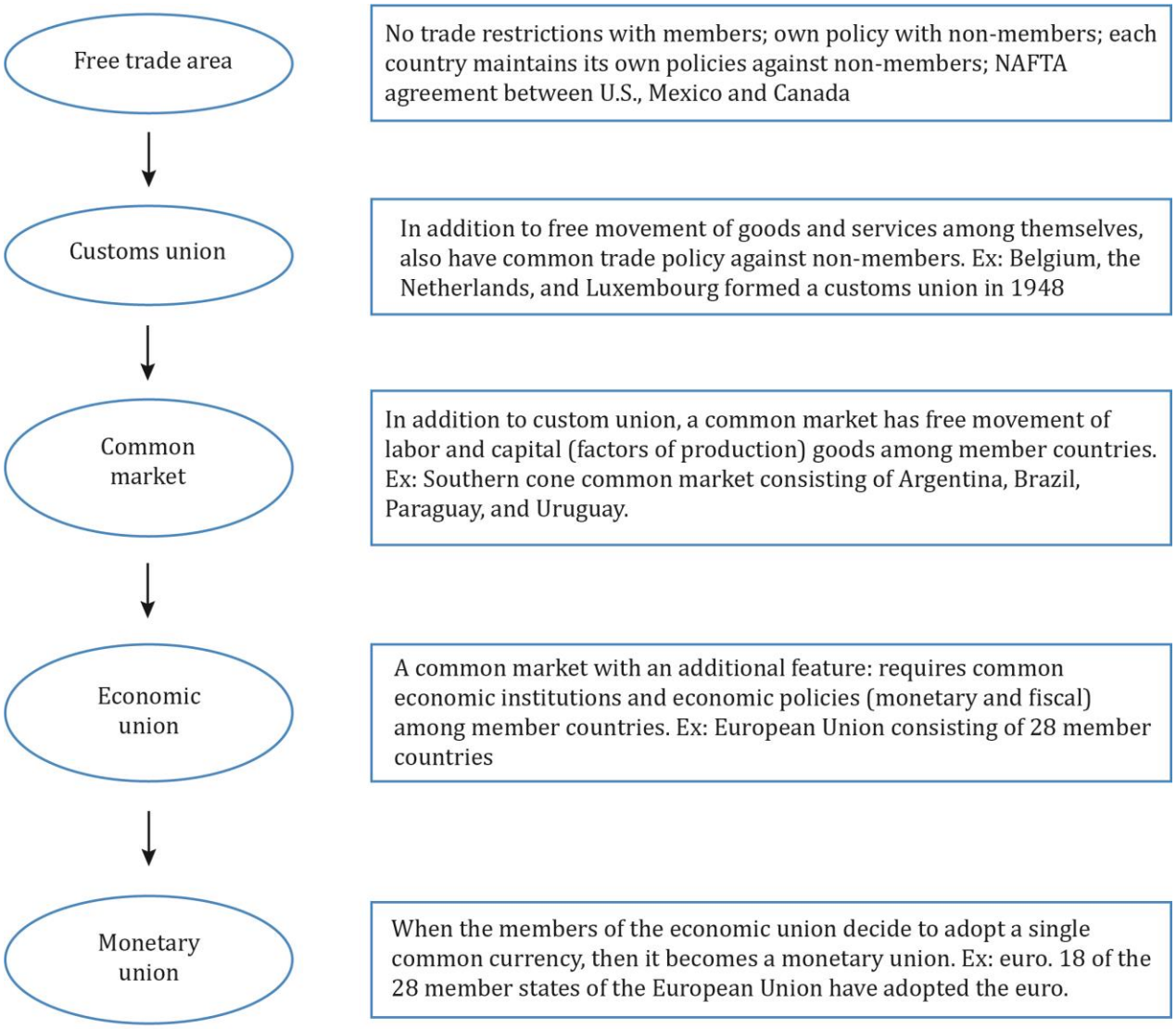
4. Trading Blocs and Regional Integration

A regional trading bloc**,** is an agreement between a group of countries, that are geographically close to each other, to reduce and eliminate barriers to trade and movement of factors of production among the members of the bloc. They have zero or very low tariffs on imports from members.

Types Of Trading Blocs

The diagram below shows the various types of regional trading blocs in an increasing order of integration.

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Regional Integration

Regional integration is popular because removing trade and investment barriers among a small group of countries is easier, politically less contentious, and faster than multilateral trade negotiations under the World Trade Organization. (WTO).

Regional integration results in preferential treatment for members over non-members and can cause changes in trade patterns.

- *Trade creation* occurs when regional integration results in the replacement of higher-cost domestic production by lower-cost imports from other members.
- *Trade diversion* occurs when lower-cost imports from non-member countries are replaced with higher-cost imports from members.

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Challenges to Deeper Integration:

There are at least two challenges to moving from an FTA to deeper levels of integration.

- Cultural and historical considerations, such as wars and conflicts, may complicate the integration process.
- A high level of economic integration limits member countries' ability to pursue independent economic and social policies.

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Summary

LO: Describe the benefits and costs of international trade.

- Benefits: Lower cost to consumers of imports due to more efficient allocation of scarce resources; higher employment, wages, and profits in export industries; economies of scale.
- Costs: Workers may be displaced and profits may be lost in industries competing with imported goods.
- Net balance: Most economists believe that the benefits outweigh the costs.

LO: Compare types of trade restrictions, such as tariffs, quotas, and export subsidies, and their economic implications.

- Tariffs: Taxes on imported goods collected by the government.
- Quotas: Limits on the absolute amount of imports allowed over some period, typically a year.
- Export subsidies: Government payments to firms that export goods which artificially reduce the cost of production.
- Voluntary Export Restraint: Agreements by exporting countries to voluntarily restrict the amount of a good that can be exported, often in the hope of avoiding tariffs or quotas imposed by trading partners.

LO: Explain motivations for and advantages of trading blocs, common markets, and economic unions.

- Free Trade Area: All barriers to import and export of goods and services among member countries are removed.
- Customs Union: In addition, all member countries adopt a common set of trade restrictions with non-members.
- Common Market: In addition, all barriers to the movement of labor and capital goods among member countries are removed.
- Economic Union: In addition, member countries establish common institutions and economic policy.
- Monetary Union: In addition, member countries adopt a single currency.

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LM07 Capital Flows and the FX Market

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2. The Foreign Exchange Market and Exchange Rates....2
3. Exchange Rate Regimes: Ideals and Historical Perspective....6
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1. Introduction

This learning module covers:

- How the foreign exchange market is structured, who the major players are, and how they conduct their business.
- How exchange rates are quoted and calculated.
- The different exchange rate regimes throughout the world.
- The effect of exchange rates on international trade and capital flows.
- Capital restrictions imposed by governments

2. The Foreign Exchange Market and Exchange Rates

Introduction and the Foreign Exchange Market

The foreign exchange market (FX) is by far the world's largest market. It has a daily turnover of about 10 to 15 times larger than global fixed income market, and about 50 times larger than global equities.

Exchange Rate

- It is the price or cost of one currency expressed in terms of another currency.
- Stated otherwise, it is the number of units of the price currency needed to buy/sell one unit of the base currency.
- Consider an exchange rate quote of 1.4500 USD/EUR.
- The numerator currency (USD) is called as the price currency and the denominator currency (EUR) is called as the base currency.
- It implies that one EUR is exchangeable with 1.45 USD.
- So, 1.4500 USD/EUR is interpreted as $1.45 per euro. Here, USD is the currency used to express price per one unit of euro.
- To determine appreciation or depreciation of a currency; if the quote rate increases in terms of the base currency → base currency has appreciated; and vice versa.
- If one currency in a currency exchange pair appreciates, the other currency depreciates.

Example

Determine whether the following currencies have appreciated or depreciated.

Currency pair	Current exchange rate	Exchange rate three years later
USD/EUR	1.4160	1.4051
INR/USD	60.4560	61.3869
CHF/USD	0.8895	0.8863

Solution:

In USD/EUR, the quoted exchange rate decreases → EUR depreciates (relative to USD), USD appreciates.

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In INR/EUR, the quoted exchange rate increases → USD appreciates (relative to INR), INR depreciates.

In CHF/USD, the quoted exchange rate decreases → USD depreciates (relative to CHF), CHF appreciates.

Nominal exchange rate is the quoted currency exchange rate at any point in time.

Real exchange rate adjusts the nominal exchange rate for inflation in each country compared to a base period.

- Real exchange rate $_{\text{price currency/base currency}}$ = nominal exchange rate $_{\text{price currency/base currency}}$ x (Base currency CPI / Price currency CPI).
- Real exchange rate shows the relative purchasing power of currencies and has the following correlations:
  - Directly related to nominal exchange rate.
  - Directly related to price level in base currency.
  - Inversely related to price level in price currency.

Example 1 from the curriculum will help understand these concepts better.

Market Participants

The forex market has a diverse range of participants. One way of classifying them is to group them based on buy-side and sell-side players.

Sell side:

- Large FX trading banks such as Deutsche bank, Citigroup, UBS, and HSBC.
- Other banks fall into the second and third tier of the FX market.

Buy side:

- Clients who use banks to undertake FX transactions.
- Corporate accounts: Corporations using forex transactions for cross-border trade of goods and services.
- Real money accounts: They usually have restricted use of leverage. Investment funds managed by mutual funds, ETFs, pension funds, endowments, etc.
- Leveraged accounts: Hedge funds, high-frequency algorithmic traders.
- Retail accounts: Individuals trading for their own accounts; tourists exchanging currency during international travel.
- Governments.
- Central banks: Intervene in the forex market to control their domestic exchange rate. For instance, during 2012-13 the Reserve Bank of India sold billions of U.S. dollars to strengthen the depreciating Indian rupee.
- Sovereign wealth funds: Countries with current account surpluses like Norway, UAE, Kuwait, and China direct international capital inflows into SWFs instead of holding

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them as foreign exchange reserves. SWF then invests these funds internationally in natural resources, infrastructure projects, and real estate to earn higher returns and exert more influence.

Market Composition

Instructor's Note: You need not memorize the numbers presented in this section.

Exhibit 2 from the curriculum (reproduced below) lists FX turnover by instrument.

Instrument	**FX Turnover (%)**
Spot	33
Outright forwards	14
Swaps[a]	49
FX options	5
Total	**100**
a Includes both FX and currency swaps.	

The largest turnover is in the swaps market, followed by the spot market.

Exhibit 3 from the curriculum (reproduced below) lists FX flows by counterparty. Average daily FX flow between financial clients is higher than that between the sell-side banks (interbank market).

Counterparty	**FX Flows (%)**
Interbank	42
Financial clients	51
Non-financial clients	8

Exhibit 4 from the curriculum (reproduced below) lists the top five currency pairs in terms of their percentage share of average daily global FX turnover.

Currency Pair	**% of Market**
USD/EUR	23.1
JPY/USD	17.8
USD/GBP	9.3
USD/AUD	5.2
CAD/USD	4.3

Exchange Rate Quotations

Exchange rate is the price of one currency relative to another. Exchange rates are typically quoted at four decimal places. The ratio or exchange rate is quoted as price currency per unit of base currency.

Consider this quote: $\frac{\text{USD}}{\text{EUR}} = 1.4000$

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It means you can buy 1.4 U.S. dollars for one euro. The currency in the denominator (one unit of the currency) is the base currency. The currency in the numerator is the price currency.

The same quote may also be written as $\frac{\text{EUR}}{\text{USD}}$ = 0.7142. If you notice, it is the reciprocal of 1.4.

Direct quote: A direct quote takes domestic currency as the price currency and the foreign currency as the base currency.

Indirect quote: An indirect quote takes domestic currency as the base currency and the foreign currency as the price currency,

Direct and indirect quotes are the reciprocal of each other.

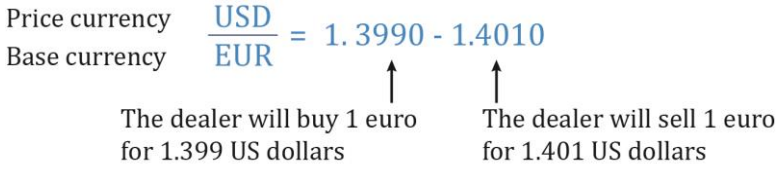
For example, from a German investor's perspective, is $\frac{\text{USD}}{\text{EUR}}$ = 1.4000 a direct quote?

The domestic currency for a German investor is the Euro. In this case, the Euro is shown as the base currency. Therefore, from the German investor's perspective this quote is an indirect quote.

Bid-ask: Currencies are always quoted as bid-ask. (*This is from the perspective of a dealer, not from the client's perspective)*. Bid rate is the rate at which the dealer will buy the base currency. Ask rate is the rate at which the dealer will sell the base currency**.**

For example**,** a bid-ask quote of $\frac{\text{USD}}{\text{EUR}}$ = 1.3990 - 1.4010 means that the dealer is willing to buy 1 euro for $1.399 and sell 1 euro for $1.4010.

The bid price is always lower than the sell price as the dealer makes money on the bid-ask spread.



Example

Appreciation of one currency is the depreciation of the other. Say the USD/EUR rate changed from 1.4 to 1.5. What is the appreciation/depreciation of each currency?

Solution:

The base currency is EUR; the price currency is USD.

The exchange rate goes up from 1.4 to 1.5. It means the base currency (EUR) has appreciated/strengthened. The USD has depreciated or weakened.

% appreciation of EUR = $\frac{1.5 - 1.4}{1.4} * 100$ = 7.142%

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To calculate the depreciation in USD, we must first convert the quote in EUR/USD terms.

Take a reciprocal of the quote to get the EUR/USD values.

Initial value: $\frac{1}{1.4}$ = 0.7143 Later value EUR/USD = $\frac{1}{1.5}$ = 0.6667

% Depreciation of dollar = $\frac{0.6667 - 0.7143}{0.7143} * 100$ = -6.67%

Note: The percentage amount by which one currency goes up (appreciates) is not necessarily the same as the percentage amount by which the other currency goes down. In our example, while the Euro appreciated by 7.142%, the U.S. dollar did not depreciate by 7.142%, instead it only depreciated by 6.67%.

3. Exchange Rate Regimes: Ideals and Historical Perspective

Every exchange rate is managed to some degree by central banks. The policy framework adopted by a central bank or the monetary authority to manage its currency relative to other currencies is called the exchange rate regime.

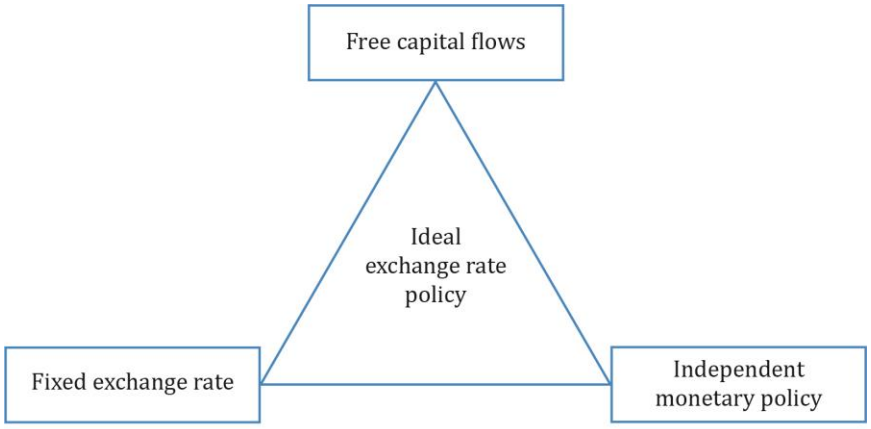
You may ask, why must the central bank intervene in the exchange rate? This is because high exchange rate volatility can affect investment decisions or affect how foreign investors perceive the investment climate (risky or stable) of a country.

Before we look at the different exchange rate regimes, let us understand what an ideal regime must have.

The Ideal Currency Regime

The ideal currency regime would have three properties:

1. The exchange rate between any two currencies would be credibly fixed.
2. All currencies would be fully convertible (i.e., currencies could be freely exchanged for any purpose and in any amount).
3. Each country would be able to undertake fully independent monetary policy in pursuit of domestic objectives, such as growth and inflation targets.



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But, in reality these three properties are not consistent. For example, if properties 1 and 2 hold good, then there would really be only one currency in the world and independent monetary policy will not be possible.

Historical Perspective on Currency Regimes

Instructor's Note: This is not part of the learning objectives and there are no practice problems. However, a brief history of the currency regime is summarized below:

- Most of the 19th century, until the start of World War 1 in 1914: the U.S. dollar and the U.K. pound sterling operated on the gold standard; meaning the price of each currency was fixed in terms of gold. Any other currency that fixed its price in terms of these two currencies was also indirectly operating on the gold standard. Trade imbalances, deflation, hyperinflation, and economies facing depression paved way for a new standard instead of gold.
- Towards the end of World War II in 1944 to the early 1970s: A Fixed parity system called "The Bretton Woods" system was introduced by John Keynes and Harry White; adopted by 44 countries replacing the gold standard. There were now fixed parities for exchange rate between currencies. There would be periodic realignments of currencies to bring them back to the fixed parity or equilibrium state. Inflation issues, countries not able to exercise monetary policy, and excessive capital mobility made countries move to the floating exchange rate system.

A Taxonomy of Currency Regimes

Exchange rate regimes for countries that do not have their own currency

Formal dollarization:

- Country uses the currency of another currency.
- The country cannot conduct its own monetary policy and imports the inflation of the country whose currency it uses.
- E.g., Panama uses the US dollar.

Monetary Union:

- Several countries use a common currency.
- Countries do not have the ability to determine their own domestic monetary policy.
- E.g., the European Union.

Exchange rate regimes for countries that have their own currency

Instructor's Note: As we move down the exchange rate regimes in the list below, the currency volatility increases and the ability to implement independent monetary policy increases.

Currency board system:

- An explicit commitment to exchange domestic currency for a specified foreign

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currency at a fixed exchange rate.
- The country cannot conduct its own monetary policy and imports the inflation of the country whose currency it uses.
- E.g., Hong Kong issues HKD only when it is backed by equivalent USD holdings.

Fixed parity:
- A country pegs its currency within margins of ± 1% vs. another currency or basket of currencies.
- It is also called as conventional fixed peg arrangement.
- Exchange rate is maintained through:
  - Direct intervention: Buying and selling foreign currency.
  - Indirect intervention: Interest rate policies or regulating foreign transactions.
- E.g., Saudi Arabia uses a fixed parity with USD.

Target zone:
- Similar to a fixed parity, but with wider bands (± 2%).
- Authorities have more flexibility in monetary policies.

Crawling peg:
- Passive crawling peg: Allows for periodical adjustments in exchange rate, typically done to adjust for higher inflation versus the currency used in the peg.
- Active crawling peg: A series of adjustments are pre-announced and implemented over time making domestic inflation predictable.
- E.g. China.

Crawling bands:
- The width of bands used in fixed peg is increased over time to make the gradual transition from fixed parity to a floating rate.

Managed float:
- Monetary authority attempts to influence the exchange rate in response to specific indicators – balance of payments, inflation rates, or employment – without any specific target exchange rate.

Independently float:
- Exchange rate is entirely market-driven.
- Interventions are used only to reduce market fluctuations.

Exchange Rates and the Trade Balance: Introduction

If a country imports more goods and services than it exports, it has a trade deficit. This deficit must be financed by borrowing from foreigners or by selling assets to foreigners. Thus, a trade deficit must be exactly matched by an offsetting capital account surplus.

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On the other hand, if a country exports more goods and services than it imports, it has a trade surplus. This surplus must be invested by lending to foreigners or by buying assets from foreigners. Thus, a trade surplus must be exactly matched by an offsetting capital account deficit.

The relationship between the trade balance and expenditure/saving decisions can be expressed as follows:

$$(X - M) = (S - I) + (T - G)$$

where:
X is exports
M is imports
S is private savings
I is investment in plant and equipment
T is taxes net of transfers
G is government expenditure

We can see from this relationship that a trade surplus (X > M) must be supported by a fiscal surplus (T > G), an excess of private saving over investment (S > I), or both.

In other words, a trade surplus indicates that the country saves more than enough to cover its investment (I) in plant and equipment. Excess savings are used to accrue financial claims against the rest of the world. A trade deficit, on the other hand, indicates that the country does not save enough to fund its investment spending (I) and must reduce its net financial claims on the rest of the world.

4. Capital Restrictions

Key points related to capital restrictions are presented below:

- Some governments restrict the inward and/or outward flow of capital.
- Restrictions on inflows might be due to strategic or defense-related reasons.
- During an economic crisis, capital might flow out of the country.
- Countries with scarce foreign exchange might restrict outflows and might also want to boost local investment.
- Over the long term, capital restrictions reduce welfare.

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Summary

LO: Describe the foreign exchange market, including its functions and participants, distinguish between nominal and real exchange rates, and calculate and interpret the percentage change in a currency relative to another currency.

Exchange rate is the number of units of the price currency needed to buy/sell one unit of the base currency. In case of an exchange rate quote of 1.4000 USD/EUR, we refer to the numerator currency (in this case, the US Dollar) as the price currency and the denominator currency (in this case, the Euro) as the base currency. This quote is a direct quote for a US-based investor and an indirect quote for a euro-based investor.

Nominal exchange rate is the quoted rate at any point in time.

Real exchange rate is the nominal exchange rate adjusted for inflation, in each country, compared to a base period. Real exchange rate $_{p/b}$ = nominal exchange rate $_{p/b}$ x (Base currency CPI / Price currency CPI).

In terms of the value of daily transactions, currency is the largest market. Within currency, FX swaps are the largest, followed by the spot market. Transactions related to speculation and hedging far exceed transactions related to international trade.

FX Market Participants:

Sell side: market makers; large multinational banks.

Buy side: corporations, investment fund managers, hedge fund managers, investors, governments and central banks.

Percentage change in a currency relative to another currency:

When USD/EUR goes down from 1.4160 to 1.4000, the USD price of a euro goes down (1.4000/1.4160) – 1 = -1.13%. We say that the euro has depreciated relative to the USD by 1.13%.

In this case the EUR/USD exchange rate has gone up from 1/1.416 = 0.704 to 1/1.4 = 0.714. The euro price of a US Dollar has gone up (0.714 / 0.704) – 1 = 1.46%. We say that the US Dollar has appreciated relative to the euro by 1.46%.

LO: Describe exchange rate regimes and explain the effects of exchange rates on countries' international trade and capital flows.

Exchange rate regimes for countries that do not have their own currency:

- Formal dollarization – A country uses the currency of another currency, typically the US dollar.
- Monetary Union – Several countries use a common currency.

Exchange rate regimes for countries that have their own currency:

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- Currency board system: An explicit commitment to exchange domestic currency for a specified foreign currency at a fixed exchange rate.
- Fixed parity: A country pegs its currency within margins of 1% vs. another currency.
- Target zone: Like fixed parity, but with wider bands (± 2%); gives monetary authority more flexibility.
- Crawling peg: Exchange rate is adjusted periodically, typically to adjust for higher inflation versus the currency used in the peg.
- Managed float: Monetary authority attempts to influence the exchange rate in response to specific indicators – balance of payments, inflation rates, or employment – without any specific target exchange rate.
- Independently float: Exchange rate is entirely market-driven.

The impact of exchange rates and other factors on a country's trade balance is mirrored by their impact on that country's capital flows.

LO: Describe common objectives of capital restrictions imposed by governments.

Key points related to capital restrictions are:

- Some governments restrict the inward and/or outward flow of capital.
- Restrictions on inflows might be due to strategic or defense-related reasons.
- During an economic crisis, capital might flow out of the country.
- Countries with scarce foreign exchange might restrict outflows and might also want to boost local investment.
- Over the long term, capital restrictions reduce welfare.

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LM08 Exchange Rate Calculations

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1. Introduction

This learning module covers:

- How to calculate cross exchange rates
- How to calculate forward exchange rates

2. Cross-Rate Calculations

Given two exchange rates and three currencies, it is possible to determine the third exchange rate. This way of determining the third exchange rate by converting one foreign exchange quote into another is called the cross rate.

Given the two exchange rates below, what is the $\frac{PKR}{INR}$ rate?

Currency	Spot rate
PKR/USD	250.0000
INR/USD	80.0000

Solution:

$\frac{PKR}{INR} = \frac{PKR}{\cancel{USD}} * \frac{\cancel{USD}}{INR}$ In this equation, the USD cancels out giving us $\frac{PKR}{INR}$.

We are given the value of $\frac{PKR}{USD}$. To get the value of $\frac{USD}{INR}$, we take the reciprocal of $\frac{INR}{USD}$ which is given.

$\frac{PKR}{INR} = 250 * \frac{1}{80} = 3.125$

Triangular arbitrage: If the implied cross rate is not equal to the quoted cross rate, then an arbitrage opportunity exists and it is called triangular arbitrage. In such cases, one would profit by buying low and selling high. For example, in the case above, if the bank quoted a rate of 3.250 for $\frac{PKR}{INR}$, then you may buy INR (sell PKR) for 3.125 and sell INR (buy PKR) for 3.250 to profit from the mispricing.

Example: Cross Rates
(*This is based on Example 1 from the curriculum.*)

Given the following exchange rates:

Currency	Spot rate
USD/EUR	1.1701
CHF/USD	0.9900
USD/GBP	1.3118

1. Calculate the spot CHF/EUR cross rate.

2. Calculate the spot GBP/EUR cross rate.

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Solution to 1:

$$\frac{\text{CHF}}{\text{EUR}} = \frac{\text{USD}}{\text{EUR}} \times \frac{\text{CHF}}{\text{USD}} = 1.1701 \times 0.9900 = 1.1584$$

Solution to 2:

$$\frac{\text{GBP}}{\text{EUR}} = \frac{\text{USD}}{\text{EUR}} \times \left(\frac{\text{USD}}{\text{GBP}}\right)^{-1} = \frac{\text{USD}}{\text{EUR}} \times \frac{\text{GBP}}{\text{USD}} = \frac{1.1701}{1.3118} = 0.8920$$

3. Forward Rate Calculations

Spot Exchange rates: Spot transactions involve the exchange of currencies for immediate delivery. For most currencies, this corresponds to "T+2" delivery, meaning that the exchange of currencies is settled two business days after the trade is agreed to by the two sides of the deal. The exchange rate used for such transactions is called the spot exchange rate.

Forward Exchange rates: Forward contracts are agreements to exchange one currency for another on a future date at an exchange rate agreed upon today. Any exchange transaction that has a settlement date later than T+2 is a forward contract. The exchange rate used for such transactions is called the forward exchange rate.

Forward exchange rates are generally quoted in terms of points or pips. The table below lists the spot rate and several forward rates for $\frac{\text{USD}}{\text{EUR}}$ currency pair. The negative sign indicates that the forward rate is less than the spot rate, i.e., the base currency EUR is weakening relative to the US dollar.

Maturity	Spot rate or forward points
Spot	1.2875
One week	- 0.3
One month	- 1.1
Three months	- 5.5
Six months	- 13.3
Twelve months	- 26.5

To convert forward points into a forward rate, divide the points by 10,000 if the exchange rate uses a four decimal place convention and by 100 if the exchange rate uses a two decimal place convention, and then add to the spot rate.

What is the 12-month forward rate?

$$\text{Forward rate} = 1.2875 + -\frac{26.5}{10{,}000} = 1.28485$$

At times, forward points can also be expressed as a percentage of the spot rate. For example,

$-\frac{0.00265}{1.2875} * 100 = -0.21\%$

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Relationship between Spot Rates, Forward Rates, and Interest Rates

The forward exchange rate depends on relative interest rates. To derive the relationship between spot rates, forward rates, and interest rates, assume you have one unit of domestic currency to be invested for let us say, one year. There are two options you may consider:

The exchange rate convention used = $\frac{\text{Foreign currency}}{\text{Domestic currency}} = \frac{\text{Price currency}}{\text{Base currency}}$

Option 1**:** Invest one unit of domestic currency (base currency) at the domestic risk-free rate i_B for one period.

Amount at the end of the period = $1+ i_B$

Option 2**:** Convert one unit of domestic currency into foreign currency (base currency) today using the spot rate $\frac{S_P}{B}$. Invest this amount at the foreign risk-free rate i_P for one period. Determine the forward rate F_P/B today at which the price currency will be converted back to base currency.

Amount of units of price currency at the end of the period **=** $\frac{S_P}{B}(1 + i_P)$

Amount in terms of base (domestic currency) at the end of the period = $\frac{S_P}{B}(1 + i_P) * \frac{1}{F_{\frac{P}{B}}}$

Instructor's Note: By using forward rate, any foreign exchange risk was eliminated in this transaction.

Since the risk of these two investments is the same, they should generate equivalent returns. As an investor, you should be indifferent between the two as it is a no-arbitrage relationship.

$1 + i_B = \frac{S_P}{B}(1 + i_P) * \frac{1}{F_{\frac{P}{B}}}$

The above equation can be rewritten as:

$$F_{\frac{P}{B}} = S_{\frac{P}{B}} * \frac{1 + i_P}{1 + i_B}$$

where:

$F_{\frac{P}{B}} = \text{forward rate}$

$i_P = \text{risk} - \text{free rate of the price currency}$

$i_B = \text{risk} - \text{free rate of the base currency}$

Let us use some values now to compute the forward rate. The spot rate for INR/USD $S_{\frac{INR}{USD}} = 100$.

The risk-free rate for INR is 10% and the risk-free rate for dollar (base currency) is 1%.

Forward rate $F_{\frac{INR}{USD}} = S_{\frac{INR}{USD}} * \frac{1 + i_{INR}}{1 + i_{USD}} = 100 * \frac{1 + 0.1}{1 + 0.01} = 108.9108$

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Some important points to be noted:

- The currency with the higher (lower) interest rate will always trade at a discount (premium) in the forward market.
- In our example above, USD was the base currency with a risk-free rate of 1%, while INR was the price currency with a risk-free rate of 10%. The forward rate calculated was 108.91 which means that the currency with the lower interest rate (USD) appreciated, while the currency with the higher interest rate (INR) depreciated. (Trading at a discount means the currency depreciates. Trading at a premium means the currency appreciates.)
- This relationship ensures that there is no arbitrage. The only forward rate that prevents arbitrage is 108.91. Otherwise, for any rate greater/lesser than 108.91 traders can exploit the mispricing by buying low and selling high.
- If the forward contract is for x days, make an adjustment based on the x/360 convention unless told otherwise. Refer to the equation below.

$$F_{\frac{P}{B}} = S_{\frac{P}{B}} * \frac{1 + (i_P * x/360)}{1 + \left(i_B * \frac{x}{360}\right)}$$

Example

Given the following data, calculate the 37-day forward rate for JPY/CAD:

Canadian dollar risk-free interest rate = 3.97%

Japanese yen risk-free interest rate = 0.23%

Spot rate $S_{\frac{JPY}{CAD}}$ = 100.87

Solution:

Forward rate = $100.87 \left(\frac{1+ 0.0023 * \frac{37}{360}}{1+ 0.0397 * \frac{37}{360}}\right)$ = 100.87 * (1.000236/1.00408) = 100.48 JPY/CAD

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Summary

LO: Calculate and interpret currency cross-rates.

The computations for cross rates are a simple case of algebraic manipulation wherein the quotes should be set up in a way that the common currency cancels out.

LO: Explain the arbitrage relationship between spot and forward exchange rates and interest rates, calculate a forward rate using points or in percentage terms, and interpret a forward discount or premium.

Arbitrage relationship between spot and forward exchange rates and interest rates:

- $F_{P/B} = S_{P/B} (1 + i_p) / (1 + i_B)$
- In this formula the interest rates have been de-annualized, if annualized rates are given, de-annualize using n/360.

Calculating a forward rate using points or in percentage terms:

- If spot rate is 1.4000 USD/EUR and the forward quote is spot plus 200 points, the forward becomes 1.4200.
- To convert points to decimal divide by 10,000: 200 / 10,000 = 0.02.
- The point is the last digit of the spot rate quote.
- We can also express the forward rate as a percentage above or below the spot rate.

Interpreting a forward discount or premium:

- If forward rate > spot rate, the base currency is said to be trading at a forward premium.
- If forward rate < spot rate, the base currency is said to be trading at a forward discount.
- The currency with the higher (lower) interest rate will always trade at a discount (premium) in the forward market.
- Interest rate differential = $i_P - i_B$
- Percentage change in spot rates is proportional to the interest rate differential.

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